Shock Syndromes I: Introduction, Vasodilatory, and Sepsis

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Learning Objectives

1. Distinguish between the various shock syndromes on the basis of a patient's clinical and hemodynamic parameters.
2. Interpret hemodynamic data from monitoring devices and markers of perfusion.
3. Devise a treatment strategy for when to use intravenous fluids and/or vasopressors in a patient with shock.
4. Develop a treatment pathway for the care of patients with sepsis or septic shock that incorporates current evidence and the Surviving Sepsis Campaign guideline recommendations.

Abbreviations in This Chapter

AVP	Arginine vasopressin
Ca_{O_2}	Arterial oxygen content
CO	Cardiac output
CVP	Central venous pressure
D_{O_2}	Oxygen delivery
Hgb	Hemoglobin
ICU	Intensive care unit
IVC	Inferior vena cava
LV	Left ventricular
LVOT VTI	Left ventricular outflow tract velocity time integral
MAP	Mean arterial pressure
O_2ER	Oxygen extraction ratio
PAC	Pulmonary artery catheter
P_{CO_2}	Partial pressure of carbon dioxide
PCWP	Pulmonary capillary wedge pressure
PH	Pulmonary hypertension
PLR	Passive leg raising (test)
PPV	Pulse pressure variation
PRBC	Packed red blood cell
PVR	Pulmonary vascular resistance
RV	Right ventricular
SBP	Systolic blood pressure
SCr	Serum creatinine
Scv_{O_2}	Central venous oxygen saturation
SOFA	Sequential Organ Failure Assessment
SSC	Surviving Sepsis Campaign
St_{O_2}	Tissue oxygen saturation
SV	Stroke volume
Sv_{O_2}	Venous oxygen saturation
SVR	Systemic vascular resistance
SVV	Stroke volume variation
V_{O_2}	Oxygen consumption

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

Questions 1 and 2 pertain to the following case.

An 80-year-old woman presents to the intensive care unit (ICU) with septic shock caused by an *Escherichia coli* urinary tract infection. Pertinent vital signs on admission are as follows: blood pressure 80/40 mm Hg, heart rate 155 beats/minute with a rhythm of atrial fibrillation, respiratory rate 26 breaths/minute, and temperature 105.8°F (41°C). On physical examination, the patient is weak, lethargic, and confused. Pertinent laboratory values are as follows: sodium (Na) 155 mEq/L, potassium (K) 3.6 mEq/L, serum creatinine (SCr) 1.8 mg/dL, and lactate 4.2 mmol/L.

1. Which clinical symptoms and physiologic variables most likely indicate that this patient has a shock syndrome?

   A. Fever, lethargy, and tachycardia.
   B. Hypotension, fever, and tachypnea.
   C. Hypotension, confusion, and hyperlactatemia.
   D. Tachypnea, fever, and hyperlactatemia.

2. Which variable is most likely contributing to impaired oxygen delivery (D_{O_2}) to the patient's end organs?

   A. Serum lactate.
   B. Atrial fibrillation.
   C. Acute kidney injury.
   D. Fever.

3. A 62-year-old woman (weight 119 kg) develops ventilator-associated pneumonia in the setting of prolonged intubation after aortic valve replacement surgery. Her pneumonia is complicated by septic shock, and she is given 2 L of 0.9% sodium chloride and 1 L of 5% albumin for resuscitation and initiated on norepinephrine. Her laboratory values are as follows: Na 144 mEq/L, chloride (Cl) 110 mEq/L, K 3.8 mEq/L, bicarbonate 18 mEq/L, SCr 1.8 mg/dL, arterial pH 7.28, and albumin 3.2 g/dL. Having

determined that the patient is still fluid responsive, you would like to give another fluid bolus. Which fluid is best for the fluid bolus?

A. 0.9% sodium chloride.
B. 5% albumin.
C. 6% hydroxyethyl starch.
D. Lactated Ringer's solution.

4. A 48-year-old man (weight 82 kg) presents to the medical ICU for septic shock secondary to a urinary tract infection. His medical history is significant only for hypertension. The patient has an initial mean arterial pressure (MAP) of 58 mm Hg and a lactate concentration of 4.8 mmol/L. The patient is initiated on broad-spectrum antimicrobials, has a central venous catheter placed in his right subclavian vein, and is resuscitated with quantitative resuscitation. He receives 2 L of 0.9% sodium chloride total and 2 L of lactated Ringer's solution, and he is initiated on norepinephrine. Three hours after presentation, his current pertinent vital signs, hemodynamic parameters, and laboratory values are as follows: MAP 68 mm Hg on norepinephrine 8 mcg/minute, central venous pressure (CVP) 10 mm Hg, central venous oxygen saturation ($Scvo_2$) 72%, urinary output 0.2 mL/kg/hour, and lactate 4.6 mmol/L. Which is the next best step for the patient's hemodynamic therapy?

A. Continue current therapy.
B. Increase the norepinephrine dose.
C. Give 1 L of 0.9% sodium chloride.
D. Initiate dobutamine.

5. An 82-year-old man is admitted to the surgical ICU after an exploratory laparotomy and small bowel resection for a small bowel obstruction that was complicated by fecal peritonitis and hypotension. The patient received 2 L of lactated Ringer's solution, 500 mL of 5% albumin, and 500 mL of 6% hydroxyethyl starch in the operating room, but vasopressors were never initiated. He remains intubated and mechanically ventilated, requiring a 90% fraction of inspired oxygen (Fio_2), with the following vital signs: heart rate 131 beats/minute in atrial fibrillation, MAP 62 mm Hg (by arterial blood pressure catheter), respiratory rate 22 breaths/minute, and temperature 100.8°F (38.2°C). An arterial blood gas show lactate 5.2 mmol/L. An arterial pulse pressure waveform analysis monitor shows pulse pressure variation (PPV) 18%. The patient has a central venous catheter in his right femoral vein. Which next step is best?

A. Give 1 L of lactated Ringer's solution.
B. Do a passive leg raising (PLR) test.
C. Measure CVP.
D. Send a blood gas for venous oxygen saturation.

6. A 19-year-old man is admitted to the medical ICU for hypotension after being stung by a bee. He was given intramuscular epinephrine by emergency medical services and transferred to the emergency department (ED). On arrival at the ED, his blood pressure was 78/42 mm Hg; he was given 1 L of 0.9% sodium chloride, diphenhydramine, famotidine, and methylprednisolone. The patient remained hypotensive but responded to an additional 2 L of 0.9% sodium chloride. He was transferred to the medical ICU for further treatment. On arrival in the medical ICU, his MAP is 62 mm Hg. A right internal jugular central venous catheter is placed, which shows CVP 3 mm Hg, $Scvo_2$ 61%, venous lactate concentration 4.4 mmol/L, and hemoglobin (Hgb) 9.6 g/dL. Together with further fluid resuscitation, which agent would be best to initiate or administer?

A. Packed red blood cells (PRBCs).
B. Dobutamine.
C. Milrinone.
D. Norepinephrine.

7. A 56-year-old man (weight 66 kg) presents to the ED with presumed community-acquired pneumonia. Blood cultures are obtained, and the patient is given ceftriaxone 1 g and levofloxacin 750 mg. His initial blood pressure is 83/47 mm Hg with a lactate concentration of 6.2 mmol/L, and he is given 1.5 L of 0.9% sodium chloride over 1 hour. Subsequently, his blood pressure is 92/54 mm Hg with a lactate concentration of 4.6 mmol/L and urinary output of 30 mL/hour. A central venous catheter placed in his right internal jugular vein shows a CVP of 6 mm Hg, and a venous blood gas reading obtained from the central venous catheter shows an $Scvo_2$ of 63%, Hgb 9.2 g/dL, and hematocrit (Hct) 28%. Which would be the best therapy for this patient right now?

A. 0.9% sodium chloride.
B. 5% albumin.
C. Phenylephrine.
D. PRBCs.

8. A 68-year-old woman (weight 88 kg) presents to the ED with a urinary tract infection. Her medical history is significant for paroxysmal atrial fibrillation. The patient's vital signs in the ED are as follows: blood pressure 83/47 mm Hg, heart rate 118 beats/minute in atrial fibrillation, respiratory rate 24 breaths/minute, and temperature 102°F (38.9°C). Her laboratory values of interest in the ED include white blood cell count (WBC) 19.6×10^3 cells/mm^3, Hgb 9.3 g/dL, albumin 2.4 g/dL, lactate 4.9 mmol/L, and SCr 1.2 mg/dL. Blood and urinary cultures are obtained, and she receives levofloxacin 750 mg and 3 L of 0.9% sodium chloride. Thirty minutes after completing the 0.9% sodium chloride infusion, her blood pressure is 92/49 mm Hg with a lactate concentration of 4.6 mmol/L and urinary output of 30 mL/hour. Which would be the best therapy for this patient right now?

   A. Norepinephrine.
   B. Vasopressin.
   C. Phenylephrine.
   D. Dopamine.

9. A 34-year-old man is admitted to the surgical ICU with septic shock associated with necrotizing soft tissue infection of the right leg. He received 100% oxygen by high-flow mask; adequate broad-spectrum antibiotics with vancomycin, piperacillin/tazobactam, and clindamycin; and quantitative resuscitation. The patient has been resuscitated for the past 4 hours with 5 L of normal saline and currently is hemodynamically unstable on norepinephrine 14 mcg/minute with a corresponding blood pressure of 92/45 mm Hg and heart rate of 132 beats/minute in sinus rhythm. His CVP is 12 mm Hg, $ScvO_2$ 72%, urinary output 0.3 mL/kg/hour, and lactate 7.4 mmol/L. Which would be best to initiate for this patient?

   A. 0.9% sodium chloride.
   B. Vasopressin.
   C. Phenylephrine.
   D. Epinephrine.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2, 4
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1, 2, 4
   b. Task B: 2
   c. Task C: 1, 3

I. INTRODUCTION

A. Shock
1. Shock is a heterogeneous group of syndromes best defined as "acute circulatory failure." This arises when the tissues receive an insufficient supply of oxygen to be able to perform vital metabolic functions.
2. Shock is often categorized into four distinct etiology mechanisms: (1) hypovolemic, (2) obstructive, (3) distributive and vasodilatory, and (4) cardiogenic. In some clinical scenarios, multiple shock syndromes can occur simultaneously.
3. The diagnosis of shock typically includes the interpretation of three variables: hemodynamic assessment, clinical presentation, and biochemical signs.
4. In many cases, shock is first identified by the presence of hypotension. However, the blood pressure limits used to define shock are arbitrary and may not be patient-specific (e.g., a patient with hypertension before critical illness).
    a. In shock states, the value typically used to describe hypotension is a systolic blood pressure (SBP) less than 90 mm Hg or a mean arterial pressure (MAP) less than 70 mm Hg.
    b. These values may vary within a range to permit autoregulation, allowing acceptable perfusion in the setting of acute hypotension.
5. Clinical presentation of shock may be subtle and can manifest in many different ways. Usually, shock is identified through an assessment of mentation, skin, and kidney function.
    a. Assessment of mentation should include a careful examination for signs of confusion and obtundation. These signs should be compared with those in the patient's preexisting status. This may be challenging in a patient who is a poor historian or in those with a diminished baseline status.
    b. Evidence of an existing shock syndrome can manifest with decreased capillary refill and cold, clammy skin.
    c. Altered kidney function in the setting of shock primarily presents with reduced urinary output (e.g., less than 0.5 mL/kg/hour). Laboratory values such as serum creatinine (SCr) often lag behind the immediate observation of urine volume and quality.
6. Biochemical assessment of patients with shock typically reveals hyperlactatemia (greater than 2 mmol/L) or reduced venous oxygen saturation (SvO_2) (less than 70%), indicating abnormal cellular oxygen metabolism.

B. Physiology
1. Hemodynamic parameters can be either directly measured from a monitoring device or calculated according to direct measurements (see Table 1).

Table 1. Hemodynamic and Oxygen Transport Values

Value	Equation (as applicable)	Normal Value
Systolic blood pressure (SBP)		90–140 mm Hg
Diastolic blood pressure (DBP)		60–90 mm Hg
Mean arterial blood pressure (MAP)[a]	[SBP + (2 × DBP)]/3	70–100 mm Hg
Heart rate (HR)		60–80 beats/min
Cardiac output (CO)[b]	HR•SV	4–7 L/min
Cardiac index (CI)	CO/BSA	2.5–4.2 L/min/m^2
Stroke volume (SV)	CO/HR	60–130 mL/beat
Pulmonary artery systolic pressure (PASP)		20–30 mm Hg

Table 1. Hemodynamic and Oxygen Transport Values *(continued)*

Value	Equation (as applicable)	Normal Value
Pulmonary artery diastolic pressure (PADP)		8–12 mm Hg
Mean pulmonary artery pressure (mPAP)	[PASP + (2 × PADP)]/3	12–15 mm Hg
Pulmonary capillary wedge pressure (PCWP) or pulmonary arterial occlusion pressure (PAOP)		5–12 mm Hg
Central venous pressure (CVP) or right atrial pressure (RAP)		2–6 mm Hg
Pulmonary vascular resistance (PVR)[c]	80 × [(mPAP − PCWP)/CO] (divide by 80 for Wood units)	20–120 dynes•s•cm-5 (< 2 Wood units)
Systemic vascular resistance (SVR)[c]	80 × [(MAP − CVP)/CO]	800–1200 dynes•s•cm-5
Oxygen delivery (Do_2)[d]	10 × CO (L/min) × Cao_2	520–570 mL/min/m^2
Arterial oxygen content (Cao_2)[d]	(1.34 × Hgb × Sao_2) + (0.003 × Pao_2)	20 mL/dL
Venous oxygen content (Cvo_2)[d]	(1.34 × Hgb × Svo_2) + (0.003 × Pvo_2)	15 mL/dL
Resting oxygen consumption (Vo_2)[d]	10 × CO (L/min) × (Cao_2 − Cvo_2)	110–160 mL/min/m^2
Oxygen extraction ratio (O2ER)	Vo_2/Do_2 × 100	20%–30%

[a]May be directly measured or calculated.

[b]May be measured using several mechanisms, including thermodilution with a pulmonary artery catheter. May also be calculated using the Fick equation (see later in the chapter).

[c]May also be expressed as an indexed value calculated by dividing the value by body surface area. BSA = body surface area in square meters.

[d]All are components of the Fick principle for cardiac output measurement.

BSA = body surface area; Cao_2 = arterial oxygen content; CI = cardiac index; CO = cardiac output; Cvo_2 = venous oxygen content; CVP = central venous pressure; DBP = diastolic blood pressure; Do_2 = oxygen delivery; Hgb = hemoglobin; HR = heart rate; MAP = mean arterial pressure; mPAP = mean pulmonary artery pressure; O2ER = oxygen extraction ratio; PADP = pulmonary artery diastolic pressure; Pao_2 = arterial partial pressure of oxygen or arterial oxygen tension; PAOP = pulmonary arterial occlusion pressure; PASP = pulmonary arterial systolic pressure; PCWP = pulmonary capillary wedge pressure; Pvo_2 = venous partial pressure of oxygen or venous oxygen tension; PVR = pulmonary vascular resistance; RAP = right atrial pressure; Sao_2 = arterial oxygen saturation; SBP = systolic blood pressure; SV = stroke volume; Svo_2 = venous oxygen saturation; Vo_2 = resting oxygen consumption.

Modified with permission from: Wittbrodt ET, Tietz KJ. Shock syndromes. In: Carter BL, Lake KD, Raebel MA, et al., eds. Pharmacotherapy Self-Assessment Program, 3rd ed. Module 2: Critical Care. Kansas City, MO: American College of Clinical Pharmacy, 1998:87-127.

2. MAP is the driving pressure for peripheral blood flow (and end-organ perfusion). Sufficient arterial pressure allows redistribution of cardiac output (CO) to vital organs.
3. Blood pressure is the product of CO and systemic vascular resistance (SVR).
4. CO is the product of heart rate and stroke volume (SV).
5. SV is determined by many factors, but predominantly preload, intrinsic contractility, and afterload.
    a. Preload refers to ventricular end-diastolic volume and is proportionally related to SV (i.e., when preload increases, the SV increases) by the Frank-Starling mechanism (though the magnitude of this relationship is reduced beyond a point of ventricular increasing end-diastolic volume).
    b. Intrinsic contractility is the ability of the myocardium to contract and may be reduced by several factors, including myocardial ischemia, cardiomyopathy, and sepsis.
    c. Afterload is the force the ventricle must overcome to eject its volume and is inversely related to SV (i.e., when afterload increases, the SV decreases). Left ventricular (LV) afterload is predominantly influenced by aortic pressure, whereas right ventricular (RV) afterload is predominantly influenced by pulmonary artery pressure.

6. SVR (also termed *total peripheral resistance*) is the resistance to flow that must be overcome by the left ventricle.
    a. SVR is the major determinant of LV afterload.
    b. Systemic vasoconstriction increases SVR, whereas vasodilation decreases SVR.
    c. Skin temperature may be used as an approximation (surrogate) of SVR, in which warm skin temperature suggests decreased SVR (vasodilation) and cold skin temperature suggests increased SVR (vasoconstriction).
7. The right ventricle better tolerates increases in ventricular volume (preload) than increases in afterload. Contrarily, the left ventricle better tolerates increases in afterload than increases in ventricular volume.
8. Coronary artery perfusion occurs primarily in diastole. Aortic diastolic pressure must be sufficient to ensure perfusion of the coronary arteries.

C. Oxygen Delivery (Do_2) and Oxygen Consumption (Vo_2)
1. The circulatory system delivers oxygen and vital nutrients to the tissue beds for homeostasis and end-organ function.
2. Oxygen is inspired and delivered to the alveoli, where it binds reversibly to hemoglobin.
3. Oxygen bound to hemoglobin is then transported by CO to the tissues. The rate of Do_2 is the product of the CO and the arterial oxygen content (Cao_2), as described in Table 1. Metabolic function of tissue beds requires consistent Do_2.
4. At the tissues, oxygen dissociates from hemoglobin and is taken up by the mitochondria through systemic capillaries for aerobic metabolism. Oxygen uptake or Vo_2 is the rate at which oxygen transfers from systemic capillaries into the tissues and is the byproduct of CO, Cao_2, and venous oxygen content (Cvo_2).
5. The Fick equation states that $CO = Vo_2/(Cao_2 - Cvo_2)$.
6. The oxygen extraction ratio (O_2ER), or the ratio of Vo_2/Do_2, is 20%–30% at resting state, meaning that about 25% of the oxygen delivered to the capillaries is taken up by the tissues. The O_2ER is relatively stable and can accommodate temporary fluctuations in Do_2 or Vo_2. Sustained Do_2/Vo_2 mismatches contribute to tissue hypoxia and deranged metabolic function.
7. Treatment of shock syndromes should be rapid to minimize permanent tissue and organ damage.
8. In the early stages of a shock state, blood pressure is preserved through stimulation of the sympathetic system, release of endogenous vasopressin, and vasoconstriction through the formation of angiotensin II. The synergy of these actions preserves blood flow and Do_2 to vital organs.
9. Blood flow is prioritized to maximize Do_2 to the heart and brain. Consequently, blood flow to extravital organs (e.g., skin, gut, kidneys) is redirected.
10. When the endogenous response to shock is insufficient either in magnitude or duration, blood pressure decreases, and overt shock develops.

II. MONITORING TECHNIQUES

A. Hemodynamic Monitoring Devices
1. Hemodynamic variables may be obtained through noninvasive or invasive monitoring devices (Table 2).
2. In patients with shock in whom the clinical examination does not lead to a clear diagnosis, further hemodynamic assessment is recommended.

3. In general, less invasive devices are desired, but they often have limited accuracy in estimating hemodynamic parameters compared with invasive devices.
4. When further hemodynamic assessment is indicated, echocardiography is the preferred modality to initially evaluate the type of shock as opposed to more invasive approaches.
5. Some ICU monitors can display additional hemodynamic parameters (e.g., automated PPV from an arterial catheter) without additional equipment or devices.
6. Bioimpedance uses electrodes on the skin or an endotracheal tube to estimate the CO through mathematical modeling of the impedance by the body to electrical current flow induced by cyclic changes in blood flow during the cardiac cycle. Bioreactance improves on bioimpedance models by analyzing the variations in the whole frequency spectra of the oscillating current instead of only variations in amplitude changes. This increases the signal-to-noise ratio and improves device performance.

Table 2. Hemodynamic Monitoring Devices

Device or Category	Obtainable Parameters	Advantages	Limitations
Noninvasive BP monitoring[a]	SBP, DBP, MAP	• Noninvasive • Bedside practitioner familiarity	• Limited accuracy in shock • Does not provide continuous monitoring • Less sensitive in predicting end-organ dysfunction
Arterial BP catheter	SBP, DBP, MAP	• More accurate BP measurement in shock than noninvasive methods • Ready access for arterial blood gas sampling • Continuous monitoring	• Invasive • Inaccurate damping influences SBP and DBP measurements (MAP still accurate) • Catheter-related infection • Brachial site lacks collateral circulation (may result in decreased distal arterial perfusion)
Central venous catheter (CVC)	CVP/RAP, $ScvO_2$	• Easier and safer to insert than a PAC • $ScvO_2$ may be available as a continuous measurement • Access for administration of highly osmotic and caustic agents	• CVP/RAP not a true estimate of LV end-diastolic pressure • CVP/RAP does not accurately predict fluid responsiveness • $ScvO_2$ not equivalent to SvO_2 (see later in the chapter)

Table 2. Hemodynamic Monitoring Devices *(continued)*

Device or Category	Obtainable Parameters	Advantages	Limitations
Pulmonary artery catheter (PAC)	Measured: PASP, PADP, mPAP, CVP/RAP, PCWP/PAOP, CO, and CI by thermodilution or continuous measurement (copper filament adapted catheter), S_{VO_2} Calculated: PVR, SVR, CO, and CI by Fick equation, SV	• Only method available to directly measure pulmonary artery pressures • Direct measurement of CO and S_{VO_2} (may be available as continuous variables)	• Outcomes data supporting superiority to CVC lacking • May cause arrhythmias • Assumes right heart function approximates left heart function (usually, but not always, true) • Fick CO calculation typically uses an estimated value for V_{O_2}, which may be falsely low in a patient with septic shock and underestimate CO • Valvular abnormalities may make values inaccurate (particularly mitral stenosis, mitral regurgitation, tricuspid regurgitation, or aortic regurgitation) • Correct catheter tip location (lung zone 3) needed for accurate readings
Echocardiography	Cardiac chamber size and function, pericardial appearance (and presence of fluid), IVC collapsibility/distensibility, ejection fraction, RVSP (an estimate of PASP), LVOT VTI (to calculate CO/CI)	• Noninvasive (transthoracic) • Visualization of ventricular function instead of presumed function based on CO • IVC collapsibility can predict fluid responsiveness	• Subjectivity of user assessment • Not done continuously; therefore, cannot detect acute changes or must be repeated when the patient's status changes • May have limited visibility/windows depending on patient body habitus or positioning
Esophageal Doppler (ODM II, CardioQ, HemoSonic 100)	CO, CI, SV, flow time	• Ease of use • May be used to determine fluid responsiveness	• Assumptions used by the device may not be valid in the setting of hemodynamic instability (fixed partition of blood flow to cephalic vessels and descending aorta, constant aortic cross-sectional area) • Accuracy depends on position (need for frequent repositioning)

Table 2. Hemodynamic Monitoring Devices *(continued)*

Device or Category	Obtainable Parameters	Advantages	Limitations
Arterial pulse pressure wave-form analysis (FloTrac/Vigileo, PiCCOplus, PulsioFlex, LiDCO plus, PRAM-MostCare, Nexfin)	CO and CI, SV, SVR (calculated), SVV, PPV	• Continuous measurement of values • Allows for assessment of SVV and PPV, which are dynamic markers of fluid responsiveness in mechanically ventilated patients (see later in the chapter) • Minimally invasive (Nexfin is noninvasive)	• Accuracy relies on optimal arterial waveform from arterial catheter • Inaccurate in patients with mitral or aortic valve disease or when used concomitantly with an intra-aortic balloon pump • Arrhythmias reduce the accuracy of reported CO and CI (though this may be accounted for by internal software with some devices) • Accuracy may be limited during rapid changes in vascular resistance • Some devices require a CVC in addition to an arterial pressure catheter • Interpretation of SVV and PPV is limited by the need for positive pressure ventilation, no spontaneous breaths being triggered by the patient, and relatively large tidal volumes • SVV and PPV are not accurate predictors of fluid responsiveness in the setting of arrhythmias
Bioimpedance/ bioreactance (NICOM, BioZ, ECOM)	Continuous CO and CI, SV, SVR, SVV	• Noninvasive • NICOM CO correlates well with CO values from thermodilution and pulse pressure waveform analysis	• Conflicting validation results with BioZ and ECOM, particularly in patients with septic shock • ECOM requires endotracheal intubation

[a]Includes manual sphygmomanometry and automated oscillometric (cuff) techniques.

BP = blood pressure; CVC = central venous catheter; CVP = cental venous pressure; IVC = inferior vena cava; LV = left ventricular; LVOT VTI = left ventricular outflow tract velocity time integral; mPAP = mean pulmonary artery pressure; PAC = pulmonary artery catheter; PADP = pulmonary artery diastolic pressure; PAOP = pulmonary artery occlusion pressure; PASP = pulmonary artery systolic pressure; PCWP = pulmonary capillary wedge pressure PPV = pulse pressure variation; RAP = right atrial pressure; RVSP = right ventricular systolic pressure; $Scvo_2$ = central venous oxygen saturation; SVR = systemic vascular resistance; SVV = stroke volume variation.

Information from: Alhashemi JA, Cecconi M, Hofer CK. Cardiac output monitoring: an integrative perspective. Crit Care 2011;15:214.

B. Markers of Perfusion
   1. Global perfusion
      a. End-organ function (altered mental status, low urinary output, and mottled skin, as noted earlier)
      b. Elevated blood lactate concentration (above 2 mmol/L)
         i. Lactate is produced from pyruvate by lactate dehydrogenase as an end product of glycolysis under anaerobic conditions.
         ii. Most lactate is cleared by the liver by conversion back to pyruvate in the Cori cycle, with a small amount cleared by the kidneys. Severe liver dysfunction may impair lactate clearance and accentuate lactate concentration elevations in shock.

iii. Elevated lactate concentrations may be the result of increased production, decreased clearance, or both.
   (a) Type A lactic acidosis occurs in the setting of Do_2/Vo_2 mismatch (oxygen demand exceeds supply).
   (b) Type B lactic acidosis is not related to tissue hypoxia and typically occurs in the setting of impaired lactate clearance or medication-related causes (e.g., metformin, epinephrine, linezolid, or toxic alcohols).

iv. Arterial and venous lactate concentrations are slightly different in value but may be used interchangeably. When trending lactate values over time, consistent methods of collection should be used.

c. Venous oximetry ($Scvo_2$ and Svo_2)

i. Central venous oxygen saturation ($Scvo_2$) and mixed venous oxygen saturation (Svo_2) are the oxyhemoglobin saturations of venous blood obtained from a central vein and the pulmonary artery, respectively, and are expressed as a percentage.

ii. $Scvo_2$ is obtained from a central venous catheter (i.e., subclavian or internal jugular access) where the catheter tip terminates in the superior vena cava. As such, $Scvo_2$ is more reflective of oxygen extraction in the brain and upper body than of systemic oxygen extraction. Of importance, the oxyhemoglobin saturation of blood obtained from a central venous catheter terminating in the IVC (i.e., femoral access) cannot be used interchangeably with the $Scvo_2$ obtained from a central venous catheter terminating in the superior vena cava and should not be used as a marker of perfusion.

iii. Svo_2 better represents systemic oxygen extraction because it represents the mixing of venous blood from the superior vena cava, inferior vena cava (IVC), and coronary sinus.

iv. In normal physiology, $Scvo_2$ is about 2%–3% lower than Svo_2 because the upper body and the brain extract more oxygen than the lower body.

v. In the setting of shock, $Scvo_2$ exceeds Svo_2 by about 5%–8% because of increased mesenteric and renal oxygen extraction with a similar cerebral extraction ratio.

vi. The difference between $Scvo_2$ and Svo_2 decreases in low CO states.

vii. Although $Scvo_2$ and Svo_2 are not equivalent, they have a good (though not perfect) correlation, and $Scvo_2$ may be a reasonable approximation of Svo_2, given its requirement for a PAC for accurate measurement.

viii. A decreased $Scvo_2$ or Svo_2 is a sign that tissue oxygen demands are not completely met by Do_2 (more discussion on this topic later in this chapter).

ix. Rearrangement of the Fick equation shows that a decrease in Svo_2 indicates a decrease in CO, whereas an increase in Svo_2 indicates an increase in CO.

x. In general, Svo_2 values above 70% are considered adequate, whereas Svo_2 values less than 40% are considered critically low and approach the critical O_2ER where anaerobic metabolism will occur and lactate concentrations will increase. Svo_2 values of 50%–70% by themselves do not lead to firm conclusions about the O_2ER and must be interpreted in the context of other markers of tissue perfusion (e.g., lactate concentrations).

xi. Svo_2 values above 80% likely indicate poor tissue oxygen extraction capacity.
   (a) This may occur because of the heterogeneity of microvascular and macrovascular blood flow (i.e., microcirculatory dysfunction), peripheral shunting of oxygen past the tissues, or impaired mitochondrial oxygen use.
   (b) Patients with septic shock and venous hyperoxia ($Scvo_2$ greater than 89%) within the first 6 hours of their treatment had a higher mortality than those with normoxia ($Scvo_2$ 71%–89%).

2. Regional tissue perfusion
    a. Microcirculatory blood flow
        i. The microcirculation consists of arterioles, capillaries, and venules and is where oxygen release to the tissues occurs.
        ii. Traditional resuscitation strategies have focused on hemodynamic and Do_2 end points (the "macrocirculation"), but the microcirculation plays a key role in tissue oxygenation in shock (particularly in septic shock) and has historically been overlooked.
        iii. Of importance, microcirculatory blood flow and Do_2 cannot be predicted by global (macrocirculatory) hemodynamics.
        iv. Microcirculatory blood flow can be visualized with orthogonal polarization spectral imaging or sidestream darkfield imaging. These devices use green light to illuminate tissue, which is absorbed by the hemoglobin of red blood cells. This allows the microcirculation to be visualized because of its red blood cell content.
        v. The sublingual microcirculation has been studied most often because of its accessibility.
        vi. Studies have shown that the microcirculation is often altered in patients with sepsis, persistent microvascular alterations are associated with multisystem organ failure and death, alterations are more severe in non-survivors than in survivors, and improvements in microcirculatory blood flow correspond with improved patient outcomes.
        vii. Decreased vascular density, decreased capillary perfusion, and a decreased percentage of perfused small vessels are the most common microcirculatory alterations. The proportion of perfused small vessels seems to be the strongest microcirculatory blood flow predictor of patient outcomes. In one study of patients with sepsis and septic shock, this was a stronger predictor of mortality than global hemodynamic markers.
        viii. Heterogeneity has been observed in microcirculatory blood flow in the same tissue bed (with as little as a few millimeters between observations) and between different tissue beds.
        ix. Evaluation of the microcirculation is not commonly used in clinical practice because it requires extensive user experience to obtain proper measurements and time to analyze the results. However, this is an attractive marker of tissue perfusion that, with technical advances, may be used more often in the future.
    b. Elevated lactate concentrations may also indicate regional tissue hypoperfusion (e.g., mesenteric ischemia or critical limb ischemia).
    c. Other measures of regional tissue perfusion include gastric tonometry to indirectly assess gastric mucosal perfusion and near-infrared spectroscopy, a noninvasive method of measuring tissue oxygen saturation (Sto_2) in a skeletal muscle.

Patient Case

1. A 77-year-old man presents to the ED with light-headedness and fatigue. He reports increasing melena during the past 24 hours. His medical history is significant for hypertension, asthma, and gastroesophageal reflux disease. Vital signs in the ED are as follows: blood pressure 88/54 mm Hg, heart rate 124 beats/minute, respiratory rate 18 breaths/minute, and temperature 102.2°F (39°C). While interviewing the patient, you note that he appears lethargic and confused. His serum chemistry panel shows the following: Na 138 mEq/L, K 3.8 mEq/L, Cl 105 mEq/L, carbon dioxide 22 mEq/L, blood urea nitrogen (BUN) 25 mg/dL, SCr 1.1 mg/dL, and glucose 78 mg/dL. Results of the complete blood cell count (CBC) are as follows: WBC 10.2×10^3 cells/mm^3, Hgb 6.6 g/dL, Hct 19.2%, and platelet count (Plt) 180,000/mm^3. Which is most likely contributing to this patient's compromised Do_2 to the end organs?
   A. Medical history of hypertension.
   B. Reduced Hgb.
   C. Tachycardia.
   D. Leukocytosis.

III. DIFFERENTIATION OF SHOCK STATES

A. Determining which shock state is present is typically based on assessments of preload (CVP or pulmonary capillary wedge pressure [PCWP]), CO ($Scvo_2$ or Svo_2 may serve as a surrogate), and afterload (SVR) (see Table 3).

B. Values to describe this hemodynamic profile have historically been obtained from a pulmonary artery catheter (PAC), but bedside echocardiography is now recommended as the preferred modality to initially evaluate the type of shock.

Table 3. Hemodynamic Profiles of Shock States

Shock State	CVP	PCWP	CO	SVR
Hypovolemic	↓[a]	↓[a]	↓	↑
Cardiogenic	↑	↑	↓[a]	↑
Obstructive				
Impaired diastolic filling (e.g., cardiac tamponade)	↑	↑	↓[a]	↑
Impaired systolic contraction (e.g., massive PE)	↑	↓ or ↔	↓[a]	↑
Vasodilatory/distributive				
Pre-resuscitation	↓	↓	↓	↓[a]
Post-resuscitation	↑	↑	↑	↓[a]

[a]Pathophysiologic hallmark of shock state.

PE = pulmonary embolism; SVR = systemic vascular resistance.

Information from: Vincent JL, De Backer D. Circulatory shock. N Engl J Med 2013;369:1726-34; Dellinger RP. Cardiovascular management of septic shock. Crit Care Med 2003;31:946-55; Weil MH, Shubin H. Proposed reclassification of shock states with special reference to distributive defects. Adv Exp Med Biol 1971;23:13-23.

C. The hemodynamic profiles in Table 3 occur when the stated shock state occurs independently, which often does not occur because patients often have features of combined shock states.

D. A patient's history of present illness can help differentiate between shock states. For example, a patient presenting after motor vehicle collision likely has hypovolemic shock, whereas a patient presenting after bee sting likely has vasodilatory/distributive shock (immune-mediated [anaphylactic] subtype).

IV. RESUSCITATION PARAMETERS AND END POINTS

A. The approach to treating a patient with circulatory shock can be divided into four phases, each having different (but sometimes overlapping) treatment goals and therapeutic strategies.
   1. The first phase focuses on salvage, in which efforts should be directed to achieving the minimum perfusion pressure and CO needed to maintain the patient's survival. Treating the underlying cause of the patient's shock, which consists of lifesaving measures, should be done at this time. Examples of these measures include antimicrobials for sepsis, revascularization for acute myocardial infarction, and surgical hemostasis for trauma.
   2. Optimization is the second phase with the goal to ensure adequate Do_2.
   3. In the third phase, patient stabilization is targeted with the goal of preventing or minimizing end-organ dysfunction.
   4. The fourth phase is de-escalation, in which the goals of therapy include vasoactive medications weaning (or cessation), fluid elimination (e.g., diuresis or ultrafiltration), and antimicrobial de-escalation as dictated by microbiology cultures, local antibiogram, and patient clinical picture.
   5. Although the rest of this chapter will focus on the first two phases, understanding the phase of a patient's circulatory shock is essential for establishing treatment goals and subsequent therapeutic approaches.

B. Blood Pressure
   1. As noted earlier, blood pressure is the driving pressure for peripheral blood flow. As such, an adequate blood pressure is vital to ensure end-organ perfusion.
   2. MAP is the true driving pressure for peripheral blood flow and end-organ perfusion and is preferred to SBP or diastolic blood pressure as a therapeutic target.
   3. The perfusion pressure of any organ can be calculated by subtracting the pressure within the organ or anatomic space from the MAP (e.g., cerebral perfusion pressure = MAP − intracranial pressure).
   4. The target blood pressure for a patient in shock is usually a MAP greater than 65 mm Hg or an SBP greater than 90 mm Hg, but this must be individualized according to other clinical/biochemical markers of perfusion.
   5. MAP is an insensitive resuscitation parameter (e.g., blood pressure may be at goal when CO is inadequate), because of this, additional resuscitation parameters should be used to ensure the optimization of all hemodynamic components that may influence end-organ perfusion and Do_2.
   6. These additional resuscitation goals typically include ensuring (1) adequate end-organ perfusion, (2) lack of fluid responsiveness, and (3) adequate Do_2.

C. Adequate End-Organ Perfusion
   1. Each organ has a critical perfusion pressure that must be exceeded to maintain adequate perfusion. This critical perfusion pressure is organ- and patient-specific because of adaptation for chronic conditions.
   2. As the MAP decreases, the perfusion pressure of the organ decreases, and subsequently, organ function decreases.
   3. Adequate organ perfusion is best assessed clinically on a per-patient basis.

4. General goals of therapy include resolution of altered mental status and adequate urinary output (above 0.5 mL/kg of body weight per hour). However, these goals may be challenging to assess in patients who are given medications that mask the ability to assess the organ function (e.g., sedatives) or in patients with chronic organ dysfunction (e.g., end-stage renal disease).

D. Lack of Fluid Responsiveness
   1. Intravenous fluids are given to increase preload and subsequently increase SV, increase CO, and increase DO_2.
   2. Fluids should be given only if there is inadequate effective organ perfusion caused by inadequate CO (presumably because of inadequate SV) and if the patient is fluid responsive.
      a. Fluid responsiveness is defined as at least a 10%–15% increase in CO after fluid administration.
      b. This is best assessed by giving a fluid challenge and evaluating the response.
      c. A change in blood pressure is not a reliable indicator of CO response to a fluid challenge; an assessment of CO (or SV) change should be used instead.
      d. In one systematic review, only 57% of hemodynamically unstable patients were fluid responsive.
   3. Patients given additional fluid when they are no longer fluid responsive will not experience the beneficial effects of fluid (increased CO)—only the harmful effects (e.g., pulmonary edema). Although an argument could be made to determine fluid responsiveness in each patient, it is most critical in patients for whom harmful fluid effects cannot be tolerated (e.g., those with refractory hypoxemia in the setting of shock).
   4. Fluid responsiveness may be predicted by static or dynamic markers.
      a. Static markers of fluid responsiveness include the cardiac filling pressures CVP and PCWP.
      b. Dynamic markers of fluid responsiveness include stroke volume variation (SVV), systolic pressure variation, pulse pressure variation (PPV), and IVC variation.
   5. Although CVP and PCWP may help differentiate shock states and may help to provide an understanding of static intravascular volume status, they are not reliable predictors of fluid responsiveness.
      a. In one study of patients with sepsis and septic shock, a CVP less than 8 mm Hg and a PCWP less than 12 mm Hg had fluid responsiveness positive predictive values of 47% and 54%, respectively.
      b. Systematic reviews and meta-analyses suggest that CVP should not be used as a resuscitation parameter.
   6. Fluid responsiveness is better predicted by dynamic markers of fluid responsiveness than by static markers of volume status or fluid responsiveness.
      a. The area under the receiver operating characteristic curve (AUC ROC), with the optimal area of 1, for predicting fluid responsiveness is as follows: PPV 0.94 (95% confidence interval [CI], 0.93–0.95), systolic pressure variation 0.86 (0.82–0.90), and SVV 0.84 (0.78–0.88). In contrast, the AUC ROC for predicting fluid responsiveness for CVP is 0.55 (0.48–0.62).
      b. Because they can be obtained from monitoring devices (Table 2), PPV and SVV are more commonly used in practice than is systolic pressure variation.
   7. The dynamic markers, PPV, SVV, and IVC collapsibility, are based on heart-lung interactions in mechanically ventilated patients.
      a. In mechanically ventilated patients, a controlled positive pressure breath increases pleural pressure. This increase in intrathoracic pressure leads to a decrease in venous return, decreased RV preload, and decreased RV SV. LV preload is subsequently decreased, which may lead to a decrease in LV SV.
      b. Patients who are preload (fluid) responsive (on the steep rather than the flat portion of the Frank-Starling curve) will have relatively large changes in LV SV with positive pressure breaths. This leads to variation in the LV SV between periods with and without positive pressure breaths.
      c. Some ICU monitors can display automated PPV from an arterial catheter without additional monitoring equipment or devices.

d. The specific values of PPV and SVV used to predict fluid responsiveness vary by study, specific conditions (e.g., use of vasopressors), and assessment method or device. In a systematic review, thresholds to predict fluid responsiveness were PPV greater than 12.5% and SVV greater than 11.6%.
e. IVC variation (also termed *IVC collapsibility* or *IVC distensibility*) uses echocardiography to visualize the diameter of the IVC during positive pressure ventilation. With a positive pressure breath from mechanical ventilation, venous return is impaired, and the diameter of the IVC increases (IVC distensibility). The opposite change in IVC diameter occurs in patients breathing spontaneously (IVC collapsibility). The change in IVC diameter during inspiration is higher in patients who are fluid responsive than in those who are not fluid responsive.
    i. In one study, the AUC ROC for predicting fluid responsiveness for IVC distensibility was 0.84 (95% confidence interval [CI], 0.63–1.0), and the best cutoff was 16% (sensitivity 67%, specificity 100%).
    ii. In a larger, multicenter study of mechanically ventilated patients with varied causes of circulatory shock, IVC distensibility had an AUC ROC for predicting fluid responsiveness of 0.64 for a threshold of 8% (sensitivity 55%, specificity 70%). In the patient subset with all assessed dynamic markers of fluid responsiveness available, the AUC ROC for predicting fluid responsiveness was significantly higher for respiratory variations in superior vena cava diameter (0.74, threshold of 21% or greater) than PPV (0.66, threshold of 11% or greater, p=0.01) and IVC distensibility (0.65, threshold of 13% or greater, p=0.02). Of note, superior vena cava diameter changes must be assessed by transesophageal echocardiography, which is not commonly performed for this purpose in the United States. IVC variation may be assessed by transesophageal or transthoracic echocardiography.

8. The passive leg raise (PLR) test measures the hemodynamic effects of a positional change in the patient's legs. Changing the position of the patient's bed such that the patient's legs are lifted to a 45-degree angle with the patient's head placed horizontally leads to a transfer of blood from the abdominal compartment and lower extremities to the intrathoracic compartment.
    a. This increase in venous return may subsequently increase SV and CO if the patient is preload responsive.
    b. An increase in CO by 10%–15% after PLR is considered a positive test.
    c. The benefit of the PLR test is that it can be used in spontaneously breathing or nonintubated patients. In addition, it does not require the administration of fluid (which may be harmful if the patient is not fluid responsive) and can easily be reversed by returning patients to their previous position.
    d. A caveat to use of the PLR test is that a method of evaluating CO is required to determine response (or lack thereof). A change in blood pressure is not an adequate surrogate marker for CO, as noted earlier. The CO measurement may be obtained from an arterial pulse pressure waveform analysis monitor (minimally invasive approach) or from a bioimpedance device, bioreactance device, or echocardiographic LVOT VTI (noninvasive approach).
9. Several caveats exist for using dynamic markers of fluid responsiveness.
    a. PPV and SVV assume the following: sinus cardiac rhythm, the absence of significant valvular dysfunction, intubation and mechanical ventilation without spontaneous breaths, and tidal volume of 8 mL/kg or more of predicted body weight. Values for PPV and SVV above the noted thresholds do not reliably predict fluid responsiveness in the setting of arrhythmias (e.g., atrial fibrillation). If these assumptions are not fulfilled, PPV and SVV are not reliable in predicting fluid responsiveness.
    b. IVC distensibility also requires intubation and mechanical ventilation without spontaneous breaths and is not conducive to continuous monitoring.
    c. The real-time response of CO (or lack thereof) with PLR must be assessed using a CO monitoring device. In addition, intra-abdominal hypertension reduces the ability of PLR to detect fluid responsiveness.

10. The best dynamic marker of fluid responsiveness to use in practice is unclear.
    a. In the previously mentioned study, the AUC ROC for predicting fluid responsiveness was higher for PPV than for SVV.
    b. In one study of postoperative patients, IVC distensibility was not noninferior to PPV (noninferiority p=0.28), considering a noninferiority margin of 15%.
    c. However, obtaining PPV from echocardiography leads to a lower AUC ROC for predicting fluid responsiveness (0.66–0.68) than does obtaining PPV from other methods (AUC ROC 0.94).
    d. In patients in whom transesophageal echocardiography is performed, superior vena cava diameter changes appear to be superior to IVC distensibility and PPV.
11. Despite the superiority of dynamic markers to static markers in predicting fluid responsiveness, incorporating dynamic markers into a resuscitation strategy that improves patient outcomes in the ICU is still lacking.
    a. In a randomized controlled trial of patients with septic shock and/or acute respiratory distress syndrome that randomized patients to treatment on the basis of pulse index continuous cardiac output (PiCCO)-derived parameters (one of the arterial pulse pressure waveform analysis devices described in Table 2) or control, 28-day mortality did not differ between groups.
    b. A pilot study using protocol-guided assessments of fluid responsiveness after initial resuscitation in patients with septic shock requiring vasopressors found this approach feasible and safe, paving the way for larger trials using this approach.

Patient Case

Questions 2 and 3 pertain to the following case.
A 59-year-old man with a medical history of cirrhosis complicated by ascites was transferred from the ward to the medical ICU for gross hematemesis, with an Hgb decrease from 9.2 g/dL to 7.3 g/dL, blood pressure 82/36 mm Hg, and new-onset confusion. After 2 L of lactated Ringer's solution and 2 units of PRBCs, the patient's Hgb increased to 9.1 g/dL, but he remained hypotensive. The medical team placed a PAC and an arterial blood pressure catheter, which showed the following: CVP 8 mm Hg, PCWP 14 mm Hg, CO 7.4 L/minute, and MAP 58 mm Hg.

2. With which shock state are the patient's post-resuscitation hemodynamic parameters most consistent?
    A. Hypovolemic.
    B. Obstructive.
    C. Vasodilatory.
    D. Cardiogenic.

3. After further resuscitation, the patient developed hypoxemia requiring intubation and mechanical ventilation. A post-intubation radiograph revealed diffuse bilateral alveolar opacities. The patient remained hypoxemic with an F_{IO_2} of 90% and was subsequently deeply sedated and given atracurium. The patient also remained hypotensive with low urinary output. Which best predicts that the patient will respond favorably to a fluid bolus?
    A. CVP 7 mm Hg.
    B. PCWP 11 mm Hg.
    C. SVV 16%.
    D. MAP 62 mm Hg.

E. Adequate Do_2
   1. CO, $Scvo_2$, and Svo_2
      a. Historically, CO was monitored and used as a therapeutic target in most patients with a PAC.
         i. Because treatment of general ICU patients with a PAC has not been shown to improve patient outcomes, routine use of a PAC has decreased substantially.
         ii. Use of venous oximetry ($Scvo_2$ and Svo_2) and echocardiographic findings has largely replaced use of a PAC.
      b. $Scvo_2$ may be used as a component of an early resuscitation strategy (though it is not a mandatory component).
         i. A decreased $Scvo_2$ or Svo_2 is a sign that tissue oxygen demands are not completely met by Do_2.
         ii. Strategies to increase Do_2 (and subsequently increase $Scvo_2$ or Svo_2) include fluids to optimize preload, red blood cell transfusion to increase Cao_2, and inotropes to increase CO. Because Pao_2 does not contribute significantly to Cao_2, it should not be used as a therapeutic target.
      c. If $Scvo_2$ or Svo_2 is used as a resuscitation goal, it may be more important to use predefined targets as well as trends in the early resuscitation period (first 6 hours after presentation) than in the later resuscitation periods.
      d. Caution must be used with using $Scvo_2$ or Svo_2 in isolation as a resuscitation goal for the following reasons:
         i. The assumption that decreased $Scvo_2$ or Svo_2 is synonymous with Do_2 and oxygen demand mismatch is not true because, by definition, tissue oxygen demand exceeds Vo_2 in shock.
         ii. During resuscitation, Vo_2 depends on Do_2, and increasing Do_2 will increase the Vo_2 without substantially changing $Scvo_2$ or Svo_2 until the critical Do_2 threshold is reached.
         iii. Svo_2 and CO are not directly proportional and are better described by a hyperbolic relationship. As such, in hyperdynamic states when the CO is already high, the Svo_2 will not increase substantially with increases in CO.
         iv. Venous hyperoxia may indicate mitochondrial dysfunction and impaired tissue oxygen use or shunting of blood from peripheral circulation; therefore, achieving a high $Scvo_2$ or Svo_2 is not always best.
      e. CO, $Scvo_2$, and Svo_2 are likely best interpreted as either adequate or inadequate (not high or low).
         i. Adequacy is best determined by assessing end-organ perfusion and lactate concentrations.
         ii. If CO, $Scvo_2$, or Svo_2 is inadequate, Do_2 should be raised.
      f. A strategy of systematically increasing CO to predefined "supranormal" values was not associated with a mortality benefit; hence, it is not recommended. The decision to augment CO must be individualized on the basis of organ perfusion.
   2. Lactate clearance and normalization
      a. Lactate clearance (a decrease in lactate concentration from the initial value) suggests improvement in global tissue perfusion and is associated with a decreased mortality rate.
      b. Significant discordance between lactate clearance and $Scvo_2$ may occur. In one study, 79% of patients with a lactate clearance of less than 10% had a concomitant $Scvo_2$ of 70% or greater.
      c. A protocol-based approach to resuscitating patients with sepsis or septic shock targeting a lactate clearance of at least 10% was noninferior to an approach targeting an $Scvo_2$ above 70%.
      d. Lactate normalization (to a concentration below 2 mmol/L) is a strong independent predictor of survival and may be an even better predictor of outcomes than lactate clearance.
      e. Targeting lactate clearance or normalization is an attractive end point because it does not require invasive hemodynamic monitoring.
      f. Use of lactate clearance together with $Scvo_2$ as a resuscitation goal may be best because this improves outcomes compared with use of $Scvo_2$ alone.
      g. Further discussion of lactate clearance as a resuscitation target in patients with sepsis or septic shock is included later in the chapter.

V. AGENTS USED TO TREAT SHOCK – FLUIDS AND VASOACTIVE AGENTS

A. See the Fluids, Electrolytes, and Nutrition chapter for a further discussion of fluid components.

B. The below description of fluids and vasoactive agents is specific to shock states in general. Description of agents used to treat septic shock will be discussed in section VII, Sepsis.

C. Pharmacology of Vasoactive Agents
   1. Vasoactive agents can be broadly differentiated to (1) vasopressors, (2) inotropes, or (3) vasodilators. Vasoactive agents may have several of these properties.
   2. Vasopressor agents are indicated if hypotension is refractory to fluid administration (the patient is no longer fluid responsive) or in the setting of severe hypotension while fluids are being administered. In addition, if a patient is severely hypotensive, vasopressors may be initiated together with additional fluid administration, even if a patient is still fluid responsive, to ensure adequate end-organ perfusion. Because there are no definitive criteria for when vasopressors should be initiated in relationship to fluid administration, bedside clinicians must often make a patient-specific assessment and decision.
   3. Vasopressor agents primarily exert pharmacologic benefit by augmenting SVR. Some vasopressors may also increase CO. Table 4 highlights the receptor pharmacology of the various agents.
      a. Once the decision to start a vasoactive agent is made, a vasoactive agent or inotrope is largely selected on the basis of the agent that best achieves the desired pharmacodynamic effect(s) (e.g., increase in SVR or increase in CO). In most shock syndromes, limited literature exists to guide optimal vasoactive agent selection.
      b. Vasopressor agents should be administered through central venous access to minimize the risk of extravasation and tissue necrosis. However, if central access is unable to be obtained, it may be necessary to administer vasopressors peripherally until central access is established. In these cases, the infusion site should be monitored closely and frequently for signs of extravasation.
      c. Catecholamine vasopressors primarily increase blood pressure by increasing SVR through their actions on α1-receptors. Differing α_1, β_1, and β_2 activity leads to differing pharmacodynamic effects between agents, as outlined in Table 4.
      d. Vasopressin increases SVR through V_1R activity in smooth muscles resulting in vasoconstriction. Activity at V_2R receptors lead to antidiuretic hormone effects in the kidney. Selective sparing of V_1R in some vascular beds or V_2R-mediated vasodilation leads to decreased vasoconstrictive effects of vasopressin in the coronary, cerebral, and pulmonary circulation.
      e. Angiotensin II also primarily increases SVR by activating AT-R1 receptors, resulting in vascular smooth muscle contraction. Activation of AT-R1 receptors also stimulates the release of norepinephrine, vasopressin, adrenocorticotropic hormone (ACTH), and aldosterone. Activation of AT-R2 receptors can counterbalance the vasoconstrictive effects of AT-R1 in some vascular beds.
      f. Some clinicians use an initial intravenous push/bolus of vasopressors ("push dose pressors") in patients with hypotension that is either severe or expected to be short in duration (e.g., medication-associated hypotension in the setting of endotracheal intubation). Common doses include phenylephrine 100–200 mcg and norepinephrine 1–2 mcg, each given every 2–3 minutes. The efficacy and safety is unclear because few studies have evaluated this approach.
   4. Inotropes exert a pharmacodynamic effect that increases CO after adequate fluid administration.
      a. Dobutamine is a β-agonist that improves cardiac function by improving SV and CO. Because of its β_1 activity, dobutamine may induce tachyarrhythmias and, given its β_2 activity, hypotension.

Table 4. Vasoactive Pharmacology and Pharmacodynamic Effects[a]

	DA	α_1	β_1	β_2	Other Mechanism	HR	CVP	CO	SVR	PVR
Vasopressors										
Dopamine[b] < 5 mcg/kg/min	++++	0	0	0		— or ↑	— or ↑	— or ↑	↔	↔
Dopamine[b] 5–15 mcg/kg/min	0	+	++++	+		↑	— or ↑	↑	— or ↑	— or ↑
Dopamine[b] > 15 mcg/kg/min	0	++++	++	++		— or ↑	— or ↑	↑	↑	↑
Epinephrine[c] < 5–10 mcg/min	0	++	++++	+++		↑	— or ↑	↑	↑	↑
Epinephrine[c] > 5–10 mcg/min	0	++++	+++	+		↑	— or ↑	↑	↑	↑
Norepinephrine	0	++++	+++	+		↓ or — or ↑	— or ↑	— or ↑	↑	↑
Phenylephrine	0	+++	0	0		— or ↓	— or ↑	— or ↓	↑	↑
Vasopressin	N/A				V_1R and V_2R agonism	— or ↓	— or ↑	↔	↑	— or ↓
Angiotensin II	N/A				AT-R1 and AT-R2 agonism	↑	— or ↑	↔	↑	?
Inotropes										
Dobutamine	0	+	++++	++		↑	— or ↓	↑	— or ↓	— or ↓
Isoproterenol	0	0	++++	++++		↑	— or ↓	↑	— or ↓	— or ↓
Levosimendan	N/A				$Ca2^+$ sensitization and KATP activation	↔	— or ↓	↑	↓	↓
Milrinone[d]	N/A				PDE3 inhibition	— or ↑	— or ↓	↑	↓	↓

[a]Receptor activity ranges from no activity (0) to maximal (++++) activity.

[b]Dopamine has dose-dependent effects. At lower doses (< 5 mcg/kg/min), activity at dopamine receptors predominates. At moderate doses (5–15 mcg/kg/min), β_1 effects predominate, and at doses > 15 mcg/kg/min, α_1 effects predominate. However, there is overlap in these effects, and these dose ranges are not exact.

[c]Epinephrine has dose-dependent effects. At lower doses (< 5–10 mcg/min), activity at β receptors predominates. At higher doses (typically > 10 mcg/min), α1 receptor activity becomes more prominent. However, there is overlap in these effects, and these dose ranges are not exact.

[d]Normal half-life is 2.5 hr, but milrinone is eliminated renally. Loading dose rarely used in routine management.

AT-R1 = angiotensin receptor type 1; AT-R2 = angiotensin receptor type 2; DA = dopaminergic; KATP = adenosine triphosphate-sensitive potassium channels; N/A = not applicable; PDE3 = phosphodiesterase type 3; V_1R = vasopressin receptor type 1; V_2R = vasopressin receptor type 2; ? = unclear.

Information from: Bangash MN, Kong ML, Pearse RM. Use of inotropes and vasopressor agents in critically ill patients. Br J Pharmacol 2012;165:2015-33; Hollenberg SM. Vasoactive drugs in circulatory shock. Am J Respir Crit Care Med 2011;183:847-55; Antonucci E, Gleeson PJ, Annoni F, et al. Angiotensin II in refractory septic shock. Shock 2017;47:560-6; Chawla LS, Busse L, Brasha-Mitchell E, et al. Intravenous angiotensin II for the treatment of high-output shock (ATHOS trial): a pilot study. Crit Care 2014;18:534; and Khanna A, English SW, Wang XS, et al. Angiotensin II for the treatment of vasodilatory shock. N Engl J Med 2017;377:419-30; Jentzer JC, Coons JC, Link CB, et al. Pharmacotherapy update on the use of vasopressors and inotropes in the intensive care unit. J Cardiovasc Pharmacol Ther 2015;20:249-60.

b. Isoproterenol is also a β-agonist, but because it has stronger β_2 activity than dobutamine, it causes more hypotension. For this reason, isoproterenol is not commonly used in the ICU as an inotrope.
c. Milrinone is a phosphodiesterase type 3 (PDE3) inhibitor. PDE3 inhibition potentiates cyclic adenosine monophosphate, leading to increased ventricular contractility and vasodilation.
    i. Milrinone may be desirable for patients receiving β-antagonists before critical illness or for patients having tachyarrhythmias while receiving dobutamine.
    ii. Milrinone may also be favored in patients with pulmonary hypertension (PH) because it decreases cardiac filling pressures and pulmonary vascular resistance (PVR).
    iii. Milrinone may increase or decrease blood pressure, depending on the individual patient's SVR and CO when it is administered. In general, a decrease in blood pressure is more common with milrinone than with dobutamine.
    iv. The half-life of milrinone is significantly longer (2.3–2.4 hours) than that of dobutamine (2 minutes) and isoproterenol (2.5–5 minutes) and thus limits its utilization in emergent scenarios.
d. Levosimendan is a calcium channel sensitizer that also activates adenosine triphosphate (ATP)-sensitive potassium channels.
    i. Levosimendan's pharmacodynamic effects are similar to those of milrinone.
    ii. Like milrinone, levosimendan may be preferred in patients receiving β-antagonists before critical illness or in those with PH.
    iii. Levosimendan is not available for use in the United States.

D. Outcomes Studies
1. Studies of therapies in patients with sepsis and septic shock will be discussed in section VII, Sepsis.
2. Fluid and resuscitation studies in ICU patients
    a. Resuscitation fluids are commonly given for patients in the ICU, with 25%–37% of patients receiving this therapy in a 24-hour period in cross-sectional studies.
    b. Crystalloids vs. colloids
        i. The Saline versus Albumin Fluid Evaluation (SAFE) study, which enrolled almost 7000 patients with varied shock types requiring fluid resuscitation, with 90% power, found no difference in 28-day mortality between treatment with 0.9% sodium chloride and 4% albumin (20.9% vs. 21.1%, p=0.87). However, this was not a study of strictly initial fluid resuscitation because the allocated study fluid was used for all fluid resuscitation in the ICU until death, discharge, or 28 days after randomization. Because of this study, crystalloids are usually preferred to albumin for the initial resuscitation of patients with shock because of their lower cost.
        ii. A pragmatic, open-label, randomized study of crystalloids compared with colloids for resuscitation found no difference between groups in 28-day mortality (27.0% vs. 25.4%, p=0.26) but did find a difference in 90-day mortality favoring the colloid group (34.2% vs. 30.2%, p=0.03). However, because 90-day mortality was a secondary (not primary) outcome, the results must be interpreted with caution. In addition, the study's open-label nature (which may bias toward finding a difference between groups) and use of many different resuscitation fluids within each study group make this study challenging to implement into practice.
    c. Balanced (chloride-poor) vs. unbalanced (chloride-rich) crystalloids
        i. Administration of chloride-rich fluids may lead to afferent renal arteriole vasoconstriction, leading to a decrease in renal perfusion and kidney injury, and may cause a metabolic acidosis by lowering the strong ion difference. As such, crystalloids that better approximate the electrolyte composition of plasma ("chloride-poor," "balanced salt," or "balanced crystalloid" solutions) have been evaluated.

Table 5. Sodium and Chloride Content of Commonly Used Resuscitation Fluids

Fluid	Sodium (mmol/L)	Chloride (mmol/L)
"Chloride-rich"[a]		
0.9% sodium chloride	154	154
5% albumin	130–160[b]	0–128[b]
Hydroxyethyl starch 6% (130/0.4)	154	154
"Chloride-poor"[a]		
25% albumin	130–160[b]	0–19[b]
Lactated Ringer's solution	130	109
Plasma-Lyte A and Plasma-Lyte 148	140	98
Normosol-R	140	98

[a]Distinction between "chloride rich" and "chloride poor" is based on chloride content > or < 120 mmol/L.

[b]Differs according to manufacturer because of differences in buffer type (e.g., sodium bicarbonate or sodium chloride) and amount used. Reported chloride content of 4% Albumex (CSL Bioplasma) is 128 mmol/L, and that of 20% Albumex (CSL Bioplasma) is 19 mmol/L (products used in Australia/New Zealand), which led to the distinction of "chloride rich" and "chloride poor" for 4%–5% albumin and 20%–25% albumin, respectively. However, neither 5% Flexbumin (Baxter, Westlake Village, CA) nor 25% Flexbumin (Baxter; products available in the United States) contains chloride.

Information from: Guidet B, Soni N, Della Roca G, et al. A balanced view of balanced solutions. Crit Care 2010;14:325; Frazee EN, Leedahl DD, Kashani KB. Key controversies in colloid and crystalloid fluid utilization. Hosp Pharm 2015;50:446-53; and Yunos NM, Bellomo R, Hegarty C, et al. Association between a chloride-liberal vs chloride-restrictive intravenous fluid administration strategy and kidney injury in critically ill adults. JAMA 2012;308:1566-72.

ii. An open-label sequential period study evaluated outcomes between a control period in which chloride-rich fluids were routinely administered (n=760) and an intervention period in which chloride-rich fluids were restricted to attending physician approval and chloride-poor fluids were routinely used (n=773). In the intervention period, the incidence of acute kidney injury (8.4% vs. 14%, p<0.001) and use of renal replacement therapy were significantly lower (6.3% vs. 10%, p=0.005). Because the study was not blinded or randomized, the results should be considered hypothesis generating.

iii. A large, double-blind, cluster randomized, double-crossover trial, the SPLIT trial, allocated four ICUs in New Zealand to either 0.9% sodium chloride (n=1025) or a balanced crystalloid solution (Plasma-Lyte; n=1067) for all patients requiring crystalloid fluid therapy. Two crossovers occurred with 7-week intervals such that each ICU used each fluid twice during the 28-week study. A total of 2278 patients were enrolled, with 2092 patients having data available for analysis of the primary outcome of the incidence of acute kidney injury within 90 days. The balanced crystalloid solution group had no lower risk of acute kidney injury than did the 0.9% sodium chloride group (9.6% vs. 9.2%, relative risk [RR] 1.04; 95% CI, 0.80–1.36; p=0.77). There was also no difference between groups in the use of renal replacement therapy (relative risk [RR] 0.96; 95% CI, 0.62–1.50; p=0.91) or hospital mortality (RR 0.88; 95% CI, 0.67–1.17; p=0.40). Of note, this study included all fluid use (not just resuscitation fluids) as well as a primarily surgical population with a relatively low severity of patient illness and patients who received relatively low volumes of fluid overall (2 L in each group).

iv. The Isotonic Solutions and Major Adverse Renal Events Trial (SMART) was a large, cluster-randomized, multiple-crossover, single-center trial, in which balanced crystalloids (lactated Ringer's solution or Plasma-Lyte A; n=7942) were compared with 0.9% sodium chloride (n=7860). The median volume of fluid received in the balanced crystalloid group was 1000 mL (interquartile range [IQR] 0–3210) and 1020 mL in the 0.9% sodium chloride group (IQR 0–3500). Patients in the balanced crystalloids group had a significantly lower incidence of a major adverse kidney event (14.3% vs. 15.4%; OR 0.90; 95% CI, 0.82–0.99; p=0.04). In-hospital mortality was not significantly different between groups (10.3% vs. 11.1%, p=0.06). Of note, in a prespecified subgroup analysis, there was a greater difference in the primary outcome favoring patients who received smaller volumes of intravenous fluids; thus, the differences between fluid types are likely magnified and more significant in patients who receive larger volumes of intravenous fluids.

v. The BaSICS study was a double-blind, randomized controlled trial that compared Plasma-Lyte 148 with 0.9% sodium chloride in ICU patients requiring fluid expansion. The intervention included all fluids over 100 mL administered for fluid challenges, maintenance fluids, and drug infusions. Over 10,000 patients were included in this study. There was no difference in the primary outcome of 90-day survival (26.4% balanced solution vs. 27.2% saline; HR 0.97; 95% CI, 0.90–1.05). Of note, this study did not evaluate fluids specifically for resuscitation; rather, it evaluated any fluid volume over 100 mL.

vi. The Plasma-Lyte 148 versus Saline (PLUS) Study randomized 2037 critically ill patients from centers cross Australia and New Zealand to receipt of either Plasma-Lyte 148 or 0.9% sodium chloride as primary fluid therapy in the ICU for 90 days. No difference was detected in the primary outcome of death from any cause within 90 days (21.8% vs. 22.0%, absolute difference −0.15; 95% CI, −3.60 to 3.30). Additionally, no difference in acute kidney development or new renal replacement therapy need was detected. Similar to the aforementioned trials, the PLUS study included all fluid use, and the intervention was not limited to resuscitation fluids.

vii. A meta-analysis of 13 randomized controlled trials (n=35,884) compared the use of balanced crystalloids with 0.9% sodium chloride and found no difference in the primary outcome of 90-day mortality (RR 0.96; 95% CI, 0.91−1.01). The Bayesian analysis determined an 89.5% probability that balanced crystalloid solutions were associated with lower mortality compared with 0.9% sodium chloride.

viii. Balanced crystalloid solutions may lead to hyponatremia (with lactated Ringer's solution) or cardiotoxicity (with acetate-containing solutions) when administered in large volumes. Caution should be used with these solutions in patients with brain injury because of the risk of cerebral edema and in those with hyperkalemia because they contain potassium (lactated Ringer solution contains 4 mmol/L of potassium, Plasma-Lyte A contains 5 mmol/L of potassium).

d. Hydroxyethyl starch solutions should not be used for fluid resuscitation in the ICU.

i. A study of 7000 critically ill patients requiring fluid resuscitation compared a low-molecular-weight, low-molar-substitution (130/0.4) hydroxyethyl starch solution with 0.9% sodium chloride. Ninety-day mortality between the hydroxyethyl starch and 0.9% sodium chloride groups did not differ (18.0% vs. 17.0%, p=0.26), but patients allocated to hydroxyethyl starch had a greater need for renal replacement therapy (7.0% vs. 5.8%, p=0.04) and a higher incidence of adverse events (5.3% vs. 2.8%, p<0.001).

ii. A systematic review and meta-analysis that analyzed only unbiased trials found an association between hydroxyethyl starch use and increased patient mortality (RR 1.09; 95% CI, 1.02–1.17; p=0.02) and need for renal replacement therapy (RR 1.32; 95% CI, 1.15–1.50; p<0.001).

3. Vasopressors and inotropes in shock
    a. A multicenter randomized trial, the SOAP II trial, included patients requiring vasopressors for shock of any type and excluded those requiring vasopressors for more than 4 hours before enrollment. Enrolled patients were allocated to either blinded norepinephrine or dopamine. Twenty-eight–day mortality between patients receiving dopamine and those receiving norepinephrine did not differ (52.5% vs. 48.5%, p=0.10), but patients receiving dopamine more often developed an arrhythmia (24.1% vs. 12.4%, p<0.001), required open-label norepinephrine (26% vs. 20%, p<0.001), and had fewer open-label vasopressor-free days (12.6 days vs. 14.2 days, p=0.007).
        i. A predefined subgroup analysis evaluated the influence of shock type on the outcome. Patients with cardiogenic shock allocated to dopamine had a higher mortality rate than those allocated to norepinephrine (log-rank p=0.03). However, the overall effect of treatment did not differ among the shock subgroups (interaction p=0.87), suggesting that the reported differences in mortality according to subgroup are spurious.
        ii. These data suggest that although norepinephrine does not improve mortality compared with dopamine, it is safer and more effective at increasing a patient's blood pressure. Given these data, a case could be made for norepinephrine as the first-line vasoactive medication of choice in all shock types.
    b. A multicenter randomized trial comparing norepinephrine with epinephrine for patients with undifferentiated shock found no difference between agents in the time to achieving a goal MAP (median 40 hours vs. 35.1 hours, p=0.26) or median number of vasopressor-free days at day 28 (25.4 days vs. 26.0 days, p=0.31). However, patients allocated to epinephrine had higher heart rates and lactic acid concentrations on the first study day (but not on subsequent days) and were more often withdrawn from the study by the treating clinician (12.9% vs. 2.8%, p=0.002). These data suggest that epinephrine has no efficacy benefits over norepinephrine and is associated with an increased incidence of adverse effects.
    c. In a systematic review and meta-analysis of vasopressors for patients with circulatory shock of all types, all-cause mortality did not differ in any comparison of different vasopressor agents or combinations.
        i. Single vasopressors evaluated included norepinephrine (reference group), dopamine, epinephrine, terlipressin, vasopressin, and phenylephrine. Vasopressor combinations included norepinephrine plus dobutamine and norepinephrine plus dopexamine (which were compared with epinephrine).
        ii. Participants treated had more arrhythmias with dopamine than with norepinephrine. The authors concluded that major changes to clinical practice are not needed but that selection of vasopressors could be better individualized and could be based on clinical variables reflecting hypoperfusion. This systematic review is limited by the small number of patients enrolled in randomized studies of some agents (e.g., phenylephrine) and few studies with combination therapy.
    d. Angiotensin II is a novel vasopressor agent being evaluated for patients with vasodilatory shock.
        i. In a phase III study of patients with shock (the Angiotensin II for the Treatment of High-Output Shock [ATHOS-3] study) (n=321) without evidence of a low CO (about 90% with sepsis), use of angiotensin II, compared with placebo, more often led to a MAP response (MAP of 75 mm Hg or greater or MAP increase of 10 mm Hg or greater, without an increase in open-label vasopressor dose) at hour 3 (69.9% vs. 23.4%, p<0.001). Angiotensin II significantly reduced the cardiovascular Sequential Organ Failure Assessment (SOFA) subscore at hour 48, but not the total SOFA score. Patients in the angiotensin II group more often developed a thrombotic event (12.9% vs. 5.1%, p=0.02), a new infection (30.1% vs. 19.0%, p=0.029), and delirium (5.5% vs. 0.6%, p=0.036).
        ii. A post hoc analysis of patients from the ATHOS-3 study with acute kidney injury requiring renal replacement therapy found that patients treated with angiotensin II had longer survival

(adjusted hazard ratio [HR] 0.44; 95% CI, 0.24–0.80) and were more likely to discontinue renal replacement therapy within 7 days (adjusted HR 2.90; 95% CI, 1.29–6.52) than those who received placebo.

iii. One retrospective study evaluating angiotensin II use found that of the 270 included patients, 67% had a positive hemodynamic response 3 hours after its initiation. Of note, 92% of patients were also receiving vasopressin at the time of angiotensin II initiation.

iv. These evaluations suggest that angiotensin II is an effective vasopressor for patients with vasodilatory shock, particularly in those with acute kidney injury, but outcomes-based studies are needed before this agent is implemented into routine clinical practice.

v. Because of the increased risk of thrombosis associated with angiotensin II, concurrent venous thromboembolism prophylaxis is recommended.

e. Levosimendan

i. In a study of patients with sepsis, adding blinded levosimendan for 24 hours compared with placebo did not lead to a significant difference in the mean SOFA score (6.68 ± 3.89 vs. 6.06 ± 3.89, respectively, p=0.053). Mortality at 28 days also did not differ between the levosimendan and placebo groups (34.5% vs. 30.9%, p=0.43). Patients in the levosimendan group more often developed a supraventricular tachyarrhythmia (3.1% vs. 0.4%, p=0.04).

ii. A study of levosimendan compared with placebo for 48 hours in patients with circulatory shock after cardiac surgery was terminated early for futility. Thirty-day mortality did not differ between the levosimendan and placebo groups (12.9% vs. 12.8%, p=0.97), nor did the groups differ in duration of mechanical ventilation or incidence of cardiac arrhythmias.

iii. These data suggest that levosimendan does not improve outcomes in patients with sepsis or circulatory shock after cardiac surgery and should not be used in these patients.

Patient Case

Questions 4 and 5 pertain to the following case.

J.B. is a 28-year-old man (weight 92 kg) who presented to the surgical ICU with shock after an appendectomy was complicated by appendiceal perforation. In the operating room, the patient received 2 L of lactated Ringer's solution and 500 mL of 5% albumin and was initiated on norepinephrine. The patient is still receiving norepinephrine 12 mcg/minute (0.13 mcg/kg/minute) with a lactate of 6.8 mmol/L and had a CO increase of 18% with a PLR.

4. Which would best meet J.B.'s fluid needs?

   A. 5% albumin.
   B. 6% hydroxyethyl starch.
   C. 0.9% sodium chloride.
   D. No fluids necessary.

5. After 12 hours in the surgical ICU, J.B. remains hypotensive with a lactate concentration of 5.2 mmol/L. He currently requires norepinephrine 14 mcg/minute (0.15 mcg/kg/minute); his MAP is 64 mm Hg and $ScvO_2$ is 61%. A bedside echocardiogram done by the ICU team reveals large ventricles with poor contractility. Which action is best?

   A. Start phenylephrine.
   B. Start vasopressin.
   C. Increase norepinephrine.
   D. Start epinephrine.

VI. VASODILATORY AND DISTRIBUTIVE SHOCK

A. Etiology and Epidemiology
   1. *Vasodilatory shock* broadly describes tissue hypoperfusion secondary to a decrease in SVR or hypoperfusion despite a normal or elevated CO, whereas *distributive shock* is technically a subset of vasodilatory shock that describes maldistribution of blood flow at the level of microcirculation (shunting) or at the organ level. However, this differentiation is likely trivial because distributive shock usually exists in vasodilatory shock, and the terms are often used interchangeably.
   2. Vasodilatory shock is the most common type of shock, with about 66% of the patients requiring vasoactive medications having this shock type.
   3. Septic shock is the most common cause of vasodilatory shock, but this shock type may also occur in several other conditions, including immune-mediated ("anaphylactic") and nonimmunologic ("anaphylactoid") reactions, neurogenic shock (classically secondary to spinal cord injury), intoxication, peridural or epidural infusion, adrenal insufficiency (Addisonian crisis), and thyroid insufficiency (myxedema coma) or as a component of ischemia-reperfusion injury (e.g., after cardiac arrest or cardiopulmonary bypass). Vasodilatory shock also occurs because of prolonged severe hypotension from any initial shock type (vasodilation is a final common pathway).
   4. The three most common causes of vasodilatory shock are septic shock (which will be covered in detail in section VII, Sepsis), immune-mediated (anaphylactic) shock, and neurogenic shock.

B. Pathophysiology
   1. Vasodilatory shock occurs because of a failure of the vascular smooth muscle cells to constrict, whether from a failure of vasoconstriction methods or the inappropriate activation of vasodilatory mechanisms. In most cases (except for neurogenic shock), this failure occurs despite high plasma concentrations of endogenous vasoconstrictors (e.g., norepinephrine, epinephrine, and angiotensin II).
   2. Potential mechanisms of vasodilation
      a. Activation of cellular ATP-dependent potassium channels leads to hyperpolarization of the vascular smooth muscle cell through potassium efflux, which prevents extracellular calcium influx by voltage-gated calcium channels. As a result, cellular depolarization is prevented, high cytosolic calcium concentrations needed for vasoconstriction are not achieved, and vasodilation occurs.
      b. Increased expression of inducible nitric oxide synthase leads to increased intracellular nitric oxide concentrations and resultant vasodilation by a cyclic guanosine monophosphate–mediated mechanism. Nitric oxide may also induce vasodilation by activating potassium channels in the plasma membrane, leading to cellular hyperpolarization, as described earlier.
      c. Inappropriately low plasma vasopressin concentrations despite the level of shock ("relative vasopressin deficiency") may contribute to the inability of the vascular smooth muscle cell to contract. Although initial plasma vasopressin concentrations may be high in the initial setting of shock, vasopressin concentrations may decrease to physiologic concentrations as quickly as 1 hour after the onset of hypotension.
   3. The pathogenesis of vasodilation depends on the underlying cause.
      a. Septic shock involves complex interactions between an infecting pathogen and the host inflammatory, immune, and coagulation response. The pattern-recognition (e.g., toll-like) receptors on innate immune system cells recognize specific molecules present in microorganisms and signal the release of nuclear factor B, which leads to the transcription of both proinflammatory cytokines (e.g., interleukin-1, interleukin-6, tumor necrosis factor alpha) and anti-inflammatory cytokines (i.e., interleukin-10). These proinflammatory cytokines activate neutrophils and endothelial cells, leading to an increased expression of inducible nitric oxide synthase and subsequent vasodilation.

b. Neurogenic shock involves a decrease in sympathetic outflow from the central nervous system with unopposed parasympathetic activity. As such, vascular tone is lost, resulting in a decrease in SVR and venous pooling of blood with a subsequent decrease in preload. Concomitant bradycardia is common, and decreased CO (even after fluid administration) may occur because of the interruption of cardiac sympathetic innervation, further contributing to hypotension. This shock type classically occurs as a complication of an acute spinal cord injury at the level of the thoracic or cervical vertebra, most commonly when the injury is above the fifth cervical vertebra.
c. Immune-mediated (anaphylactic) shock occurs because of reexposure to a sensitizing foreign pathogen that stimulates immunoglobulin E–mediated mast cell or basophil degranulation and resultant cytokine (e.g., histamine and tryptase) release. The mechanism of vasodilation is complex and multifaceted; however, for example, the binding of histamine to the histamine-1 receptor can activate nitric oxide synthase with resultant increases in nitric oxide and vasodilation.

4. Profound vasodilation leads to ineffective circulating plasma volume (either from venodilation ["venous pooling"] or from fluid shifts because of increased vascular permeability) and resultant decreases in cardiac preload and CO.

C. Resuscitation and Treatment
1. Resuscitation and treatment of patients specifically with sepsis and septic shock is covered later in the chapter.
2. The underlying cause of the shock state must quickly be addressed when resuscitation is initiated.
   a. Septic shock requires rapid (within 1 hour of disease recognition) administration of antimicrobials with activity against all likely pathogens.
   b. The potential offending agent should be discontinued (for medication-related reactions) and potentially removed (in the setting of envenomation) for patients with immune-mediated (anaphylactic) shock.
3. Treatment goals and end points of resuscitation are usually similar to those listed in section IV, Resuscitation Parameters and End Points. In the setting of acute spinal cord injury, a MAP goal of at least 85 mm Hg has been associated with improved outcomes in uncontrolled studies. As such, the guidelines recommend an initial MAP goal of 85–90 mm Hg in patients with acute spinal cord injury for 1 week after injury (level III recommendation).
4. Initial resuscitation
   a. The treatment of choice is intravenous fluids, which restore effective circulating volume.
      i. Crystalloids (e.g., lactated Ringer solution or normal saline) are typically the initial fluid of choice compared with colloids (e.g., albumin or other blood products).
      ii. Fluid should be administered until the patient is no longer fluid responsive.
   b. Vasopressors should be initiated for hypotension unresponsive to fluid administration.
      i. Norepinephrine is usually considered the first-line vasopressor because of its ability to increase SVR without decreasing CO.

ii. Epinephrine is typically used for patients with immune-mediated (anaphylactic) shock to provide mast cell stabilization, histamine suppression, bronchodilation, and vasoconstriction. During initial treatment, intramuscular epinephrine (0.01 mg/kg; maximum 0.5 mg) into the anterolateral thigh is preferred for speed and fewer adverse events, including overdoses, compared with intravenous epinephrine. Intramuscular epinephrine may be repeated at 5- to 15-minute intervals if the response is inadequate. Beyond initial management, in patients who continue to be hypotensive despite intramuscular epinephrine, epinephrine infusions (1–10 mcg/minute) are often used for ongoing shock, largely because of convention rather than evidence. In patients with actual or impending cardiac arrest, slow, intravenous administration of epinephrine 50 mcg may be necessary. Norepinephrine, vasopressin, and other vasopressor agents have been used with refractory hypotension associated with anaphylaxis, though it is unclear whether this strategy is superior to epinephrine alone. Dosing and route of administration of epinephrine for anaphalaxis should be monitored closely and verified; accidental overdoses involving decimal errors or confusing "mcg" with "mg" have led to adverse events such as cardiac arrest.

iii. In neurogenic shock, agents with combined vasoconstrictive and inotropic properties (e.g., norepinephrine, dopamine, epinephrine) are preferred. Phenylephrine may also be used, but a concomitant inotropic agent (e.g., dobutamine) is often administered.
   (a) Atropine may also be given for symptomatic bradycardia.
   (b) Adjunctive oral pseudoephedrine or midodrine may also be used to maintain a MAP greater than 85 mm Hg for 1 week.

5. Additional therapies
   a. Immune-mediated (anaphylactic) shock is also conventionally treated with concomitant histamine-1 and histamine-2 receptor antagonists and corticosteroids, though no strong data support the use of these agents. Corticosteroids have a slow onset of action (4–6 hours) and are therefore ineffective for acute management; however, they may decrease the risk of late symptoms. If used, methylprednisolone 1–2 mg/kg or the equivalent is recommended.
   b. The most recent guidelines regarding the treatment of patients with acute cervical spine and spinal cord injuries recommend against administering methylprednisolone (level I recommendation).
   c. Patients with vasodilatory shock secondary to adrenal insufficiency (Addisonian crisis) should receive intravenous corticosteroids (typically hydrocortisone 50 mg every 6 hours).
   d. Management of vasodilatory shock secondary to thyroid insufficiency (myxedema coma) is outlined in the Hepatic Failure/GI/Endocrine Emergencies chapter.

VII. SEPSIS

A. Introduction
   1. Sepsis is caused by a dysregulated host response to infection.
   2. While continuing to evolve, the diagnosis of sepsis traditionally has required a known or suspected source of infection with two or more criteria of the systemic inflammatory response syndrome. Sepsis was further classified as sepsis, severe sepsis (sepsis with organ dysfunction), or septic shock (sepsis with arterial hypotension unresponsive to fluid administration).

B. Definitions
   1. The Surviving Sepsis Campaign (SSC), a joint collaboration between the Society of Critical Care Medicine and the European Society of Intensive Care Medicine, has published five iterations of international guidelines for the treatment of patients with sepsis and septic shock. The 2021 SSC guidelines were sponsored or endorsed by 24 international organizations and inform many of the recommendations in this section.
   2. Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection.
   3. Organ dysfunction can be identified as an acute change in the total SOFA score of 2 points or higher consequent to the infection.
   4. A prompt identification tool ("qSOFA") for bedside use outside the ICU was also proposed in 2016, which includes altered mental status, SBP of 100 mm Hg or less, and respiratory rate of 22 breaths/minute or greater. Fulfillment of two of these three criteria had predictive validity for mortality similar to the full SOFA score outside the ICU (technically, a Glasgow Coma Scale score of 13 or less was used in the regression model but was simplified to any alteration in mental status for the qSOFA).
      a. Of importance, qSOFA does not define sepsis, but two or more qSOFA criteria are a predictor of both increased mortality and ICU stays of more than 3 days in non-ICU patients. These criteria should also be used to investigate further for infection.
      b. The full SOFA score is superior to the qSOFA for predicting mortality in patients in the ICU.
      c. Emerging data analyses suggest that in patients outside the ICU, the qSOFA score has a lower discriminant ability for predicting ICU admission or death than other early warning scores (NEWS or MEWS) and should be further investigated before implementation as a risk-stratification tool for patients with suspected infection.
      d. Because data are lacking to support its use as a screening tool, the 2021 SSC guidelines recommend against using qSOFA as a single screening tool for sepsis or septic shock.
   5. Septic shock is defined as a subset of sepsis in which underlying circulatory and cellular/metabolic abnormalities are sufficient to increase mortality substantially.
   6. Patients with septic shock can be identified as those with sepsis with persistent hypotension requiring vasopressors to maintain a MAP of 65 mm Hg and greater and having a serum lactate concentration greater than 2 mmol/L despite adequate fluid resuscitation.
   7. With this new definition, the systemic inflammatory response syndrome criteria are no longer used, and severe sepsis no longer exists as a clinical entity.
   8. The new sepsis definition better identifies patients at risk of in-hospital mortality secondary to sepsis (about 10% mortality) than the previous definition.
   9. These definitions will likely shift patient identification and terminology in clinical practice, particularly for patient inclusion in studies.
   10. The SSC bundles are no longer included in the SSC guidelines, but they are published on the SSC website (www.survivingsepsis.org). This website also contains the most up-to-date recommendations, tools for implementation, and official statements from the campaign regarding ongoing events (e.g., the new sepsis definitions).

C. Epidemiology and Social Determinants of Health
   1. Sepsis is common, with an annual incidence in the United States above 890,000, rising by about 13% per year, and an associated cost exceeding $24 billion.
   2. In a European epidemiologic study, 37% of all ICU patients had sepsis.
   3. Septic shock is the most common shock type, accounting for 62% of all cases of shock syndromes requiring vasopressors.
   4. In-hospital mortality associated with severe sepsis and septic shock (using previous definitions) is around 18%–25%, whereas mortality at 2 years is around 45%.

5. In-hospital mortality of patients with septic shock defined according to the new definition was 35%–54% in large retrospective data sets.
6. Many disparities exist with sepsis incidence, complications, and mortality in at-risk, minority, and underserved communities and populations.
    a. American Indians and Alaskan Natives are 1.6 times more likely to die from sepsis than all other races in the United States.
    b. Asian and Pacific Islander patients are 16% more likely to die from sepsis than White patients.
    c. Black patients are 1.7 times more likely to develop severe sepsis than White patients.
    d. Hispanic patients are 10% more likely to develop severe sepsis than White patients.

D. Pathophysiology
1. Sepsis and septic shock involve complex interactions between an infecting pathogen and the host inflammatory, immune, and coagulation response.
2. Hypotension develops through the inappropriate activation of vasodilatory mechanisms (increased nitric oxide synthesis) and the failure of vasoconstrictive pathways (activation of ATP-dependent potassium channels in vascular smooth muscle cells and vasopressin deficiency), resulting in a vasodilatory shock. In addition, blood flow is inappropriately dispersed at the organ level or in the microcirculation (shunting), leading to distributive shock.
3. Vascular endothelial cell injury leads to capillary fluid leak and a resultant decrease in preload. Venous dilation further exacerbates the decrease in cardiac preload.
4. Tissue Do_2 may further be impaired by a decrease in CO, inadequate Cao_2 (low hemoglobin concentration or saturation), or impaired oxygen unloading from hemoglobin. These multifactorial hemodynamic abnormalities lead to decreased effective tissue perfusion, in which Vo_2 exceeds Do_2 and cellular injury results. This further compounds the proinflammatory and procoagulant state, precipitating multiple organ dysfunction and possibly death.

E. Diagnosis
1. There is no specific diagnostic test for sepsis or septic shock, and the diagnosis is typically based on the definitions noted earlier.
2. Recognizing that pneumonia is the most common site of infection (in about 45% of cases), followed by intra-abdominal infection (about 30%) and urinary tract infection (about 11%), microbial cultures should be sent from all suspected infectious sites as soon as possible.
3. All microbial cultures are negative in more than 25% of cases, and only 30% of patients have positive blood cultures. The most commonly isolated pathogens in ICU patients are gram-negative bacteria (62%), gram-positive bacteria (47%), and fungi (19%).
4. Cultures should be obtained before antimicrobial therapy is initiated unless doing so would significantly delay (greater than 45 minutes) therapy.

F. Treatment of Sepsis and Septic Shock
1. The pharmacologic-related 2021 SSC recommendations can be found in Box 1.

Box 1. Pharmacologic-Related Surviving Sepsis Campaign Recommendations for Patients with Sepsis and Septic Shock[a]

Initial Resuscitation
1. Sepsis and septic shock are medical emergencies, and treatment and resuscitation are recommended to begin immediately (BPS)
2. For patients with sepsis-induced hypoperfusion or septic shock, at least 30 mL/kg of IV crystalloid fluid is suggested to be given within the first 3 hr of resuscitation (weak recommendation, low quality of evidence)
3. Dynamic measures to guide fluid resuscitation are suggested over physical examination or static parameters alone (weak recommendation, very low quality of evidence)
4. Guiding resuscitation to decrease serum lactate in patients with elevated lactate level is suggested (weak recommendation, low quality of evidence)
5. Capillary refill time to guide resuscitation as an adjunct to other measures of perfusion is suggested (weak recommendation, low quality of evidence)
6. An initial target MAP of 65 mm Hg is recommended in patients with septic shock requiring vasopressors (strong recommendation, moderate quality of evidence)
Antimicrobial Therapy
1. In patients with possible septic shock or a high likelihood of sepsis, administering antimicrobials immediately, ideally within 1 hr of recognition, is recommended (strong recommendation, low quality of evidence for septic shock; very low quality of evidence for sepsis without shock)
2. In patients with possible sepsis without shock, rapid assessment of the likelihood of infectious vs. noninfectious causes of acute illness is recommended (BPS)
3. A time-limited course of rapid investigation and, if concern for infection persists, the administration of antimicrobials within 3 hr from the time when sepsis was first recognized is suggested for patients with possible sepsis without shock (weak recommendation, very low quality of evidence)
4. In patients with a low likelihood of infection and without shock, deferring antimicrobials while continuing to closely monitor the patient is suggested (weak recommendation, very low quality of evidence)
5. Optimizing dosing strategies of antimicrobials based on accepted pharmacokinetic/pharmacodynamic principles and specific drug properties is recommended (BPS)
6. Empiric antimicrobials with MRSA coverage are recommended in patients with sepsis or septic shock at high risk of MRSA (BPS)
7. Two antimicrobials with gram-negative coverage are suggested for empiric treatment in patients with sepsis or septic shock and at high risk for MDR organisms (weak recommendation, very low quality of evidence)
8. Double gram-negative coverage is not suggested once the causative pathogen and susceptibilities are known (weak recommendation, very low quality of evidence)
9. Empiric antifungal therapy is suggested in patients with sepsis or septic shock at high risk of fungal infection (weak recommendation, low quality of evidence)
10. Daily assessment for de-escalation of antimicrobials is suggested over using fixed durations of therapy (weak recommendation, very low quality of evidence)
11. In patients with an initial diagnosis of sepsis or septic shock and adequate source control, shorter over longer durations of antimicrobial therapy are suggested (weak recommendation, very low quality of evidence)
12. Procalcitonin in combination with clinical evaluation is suggested to decide when to discontinue antimicrobials in patients with an initial diagnosis of sepsis or septic shock and adequate source control where optimal duration of therapy is unclear (weak recommendation, low quality of evidence)

Box 1. Pharmacologic-Related Surviving Sepsis Campaign Recommendations for Patients with Sepsis and Septic Shock[a] *(continued)*

Fluid Therapy
1. Crystalloids are recommended as first-line fluid for resuscitation (strong recommendation, moderate quality of evidence)
2. Balanced crystalloids instead of normal saline for resuscitation are suggested (weak recommendation, low quality of evidence)
3. Albumin is suggested in patients who received large volumes of crystalloids (weak recommendation, moderate quality of evidence)
4. Starches are not recommended for resuscitation (strong recommendation, high quality of evidence)
5. Gelatin is not suggested for resuscitation (weak recommendation, moderate quality of evidence)
Vasoactive Medications
1. Norepinephrine is recommended as the first-line vasopressor (strong recommendation, high quality of evidence over dopamine, moderate quality of evidence over vasopressin, low quality of evidence over epinephrine)
2. Vasopressin is suggested in patients with septic shock on norepinephrine with inadequate MAP levels, instead of escalating the dose of norepinephrine (weak recommendation, moderate quality of evidence)
3. Epinephrine is suggested in patients with septic shock and inadequate MAP levels despite norepinephrine and vasopressin (weak recommendation, low quality of evidence)
4. For patients with septic shock and cardiac dysfunction with persistent hypoperfusion despite adequate volume status and arterial blood pressure, either adding dobutamine to norepinephrine or using epinephrine alone is suggested (weak recommendation, low quality of evidence)
5. Adding dobutamine to norepinephrine or using epinephrine alone is suggested in patients with septic shock and cardiac dysfunction with persistent hypoperfusion despite adequate volume status and arterial blood pressure (weak recommendation, low quality of evidence)
6. Starting vasopressors peripherally to restore MAP rather than delaying initiation until a central venous access is secured is suggested (weak recommendation, very low quality of evidence)
Corticosteroids
1. Corticosteroids are suggested for patients with septic shock and an ongoing requirement for vasopressor therapy (weak recommendation, moderate quality of evidence)

[a]Strength and evidence level of recommendations listed in parentheses are according to the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system. The system classifies strength of recommendations as strong or weak and quality of evidence as high, moderate, low, or very low. BPSs represent ungraded strong recommendations developed using strict criteria.

BPS = best practice statement; IV = intravenous(ly); MAP = mean arterial pressure; MDR = multidrug resistant; MRSA = methicillin-resistant *Staphylococcus aureus*.

Information from: Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. Crit Care Med 2021;49:e1063-e1143.

2. Early recognition and treatment of patients with sepsis, particularly those with sepsis-induced end-organ hypoperfusion, is of utmost importance.
3. Initial resuscitation
    a. Initial goals of therapy are to restore effective tissue perfusion (by administering intravenous fluids and vasoactive medications) while treating the underlying cause of the syndrome (through antimicrobial administration and infectious source control, as applicable).
    b. Patients should first be administered a fluid challenge of 30 mL/kg of crystalloid solution as quickly as possible.
    c. Quantitative resuscitation is no longer recommended for routine use in patients with sepsis or septic shock.
        i. This strategy includes intensive monitoring (e.g., placement of a central venous [superior vena cava] catheter or echocardiography), setting goals for hemodynamic support, and using therapies to achieve those goals (e.g., use and optimization of fluids, vasopressors, and Do_2 methods).
        ii. Instead, further resuscitation should be guided by reassessment of hemodynamic parameters.
    d. Quantitative resuscitation strategies have been evaluated in several large studies.
        i. "Early goal-directed therapy"
            (a) The landmark study of early goal-directed therapy evaluated 263 patients with sepsis and septic shock treated in the ED for the first 6 hours after presentation.
            (b) Treatment of patients in the standard therapy group was at the clinician's discretion, whereas treatment of those in the intervention arm uniformly incorporated a protocol to achieve the previously mentioned goals and incorporated $Scvo_2$ as a treatment goal (70% or greater).
            (c) Early, aggressive, goal-directed resuscitation was associated with a 16% absolute risk reduction in hospital mortality compared with standard therapy (30.5% vs. 46.5%, $p<0.009$).
            (d) Critiques of this study include a higher-than-expected mortality rate in the standard therapy arm, the incorporation of CVP as a resuscitation goal, and the study's single-centered nature.
            (e) Many centers faced logistical and financial barriers to implementing the early goal-directed therapy protocol. As such, alternative approaches to quantitative resuscitation were developed and studied.
        ii. Protocolized Care for Early Septic Shock (ProCESS) study
            (a) Randomized 1341 ED patients with septic shock in U.S. academic medical centers to three treatment arms:
                (1) Early goal-directed therapy (with the same protocol as noted previously)
                (2) Protocol-based standard care that did not require use of a central venous catheter but that used clinician judgment for fluid administration and hypoperfusion, together with a shock index (heart rate/SBP) of less than 0.8 and an SBP greater than 100 mm Hg as resuscitation targets
                (3) Usual care (treatment according to the bedside physician)
            (b) 60-day mortality did not differ between the treatment arms (early goal-directed therapy 21.0% vs. protocol-based standard care 18.2% vs. usual care 18.9%, p=0.55, for the three-group comparison).
            (c) Patients in the early goal-directed therapy arm were more often admitted to the ICU (91.3% vs. protocol-based standard care 85.4% vs. usual care 86.2%, p=0.01).
            (d) This study has several critiques, primarily a likely shift of usual care toward therapy that resembles quantitative resuscitation (both of which may have contributed to a type II error).

iii. Australasian Resuscitation in Sepsis Evaluation (ARISE) study
   (a) Randomized ED patients in both academic and nonacademic centers mainly in Australia and New Zealand to either early goal-directed therapy (with the same protocol noted previously) or usual care (treatment according to the bedside physician)
   (b) 90-day mortality did not differ between the treatment arms (early goal-directed therapy 18.6% vs. usual care 18.8%, p=0.90).
   (c) Because the enrolled patients had lower severity-of-illness scores and lower mortality at 90 days, they may have been less acutely ill than the patients enrolled in the ProCESS trial. However, about 70% of the patients in the ARISE study had septic shock at randomization (only 54% of the patients in the ProCESS trial met this criterion), suggesting the patients in ARISE were critically ill and the intended population was studied.

iv. Protocolised Management in Sepsis (ProMISe) study
   (a) Randomized 1260 ED patients in both academic and nonacademic centers in England to either early goal-directed therapy or usual care (identical to the ARISE study in design)
   (b) 90-day mortality did not differ between those treated with early goal-directed therapy and those treated with usual care (29.5% vs. 29.2%, p=0.90).
   (c) Patients in the early goal-directed therapy arm had higher SOFA scores at 6 hours (mean 6.4 ± 3.8 vs. 5.6 ± 3.8, p<0.001), more often received advanced cardiovascular support (37.0% vs. 30.9%, p=0.026), and had a longer ICU length of stay (median [IQR] 2.6 [1.0–5.8] days vs. 2.2 [0.0–5.3] days, p=0.005).

v. Discussion of ProCESS, ARISE, and ProMISe studies
   (a) In a systematic review and meta-analysis that included the ProCESS, ARISE, and ProMISe studies, early goal-directed therapy was not associated with a difference in mortality compared with control (OR 1.01; 95% CI, 0.88–1.16; p=0.9), but it was associated with increased vasopressor use (OR 1.25; 95% CI, 1.10–1.41; p<0.001) and more frequent ICU admission (OR 2.19; 95% CI, 1.82–2.65; p<0.001).
   (b) Patients in these studies received about 30 mL/kg of crystalloid solution before study enrollment. This is significantly different from the patients in the landmark early goal-directed therapy study, in which patients were enrolled before resuscitation.
   (c) The ARISE and ProMISe studies required antimicrobial administration before enrollment. In ProCESS, 76% of patients had antimicrobials administered before enrollment and 97% within 6 hours of enrollment. In the landmark early goal-directed therapy study, only 86% of patients had antimicrobials administered within 6 hours of enrollment.
   (d) These studies highlight the benefits of timely administration of antibiotics and intravenous fluids, which should be a focus of the early care of patients with sepsis and septic shock.
   (e) 51%–62% of patients in the usual care arms of these studies had a central venous catheter inserted, even though it was not required in the study protocol.
   (f) The consistent findings of lower mortality rates in the contemporary studies suggest that care of patients with septic shock has evolved since the landmark early goal-directed therapy study.
   (g) In centers with ubiquitous early recognition and aggressive resuscitation strategies, protocolized care may no longer be mandatory.
   (h) Restrictive fluid approach
      (1) Recent studies have evaluated approaches of restricting early fluid resuscitation in patients with septic shock.

(2) The Crystalloid Liberal or Vasopressors Early Resuscitation in Sepsis (CLOVERS) trial compared clinical outcomes associated with a restrictive fluid strategy with early vasopressors to a liberal fluid strategy (consisting of 2 L crystalloid followed by additional boluses triggered by clinical markers). There was no detected difference in death before discharge home by 90 days between therapeutic strategies (absolute difference –0.9 [95% CI –4.4 to 2.6]) or other clinical outcomes.

(3) The 2022 CLASSIC trial randomized 1554 patients with septic shock who had received at least 1 L of intravenous fluids to either a restrictive-fluid group or a standard-fluid group in which fluid administration was protocolized and restricted to 250–500 mL of crystalloids. No difference in death by 90 days was seen between intervention arms (42.3% vs 42.1%; RR 1.00; 95% CI, 0.89–1.13) or serious adverse events (29.4% vs. 30.8%; RR 0.95; 95% CI, 0.77–1.15).

vi. Lactate clearance

(a) In one study of patients with sepsis and septic shock, failure to achieve a lactate clearance of at least 10% was associated with a higher mortality rate than was failure to achieve an $Scvo_2$ greater than 70%.

(b) A large, multicenter, noninferiority study randomized patients to a quantitative resuscitation protocol that incorporated $Scvo_2$ as a treatment goal or to a quantitative resuscitation protocol that incorporated a lactate clearance of 10% or greater as a treatment goal.

(1) The study aimed to address whether using $Scvo_2$ as a treatment goal was necessary in a quantitative resuscitation protocol.

(2) The lactate clearance strategy was noninferior to the $Scvo_2$ strategy (in-hospital mortality 17% vs. 23%, respectively; 95% CI for mortality difference -3% to 15% [above the -10% predefined noninferiority threshold]).

(3) Only 10% of the patients in each arm received a therapy specifically directed to improve either lactate clearance or $Scvo_2$, which biased the study toward finding no difference between treatment arms (and incorrect rejection of a true null hypothesis of the existence of a difference between groups in this noninferiority study; a type I error).

(4) Equally pertinent is that most patients with septic shock in this study (90%) achieved adequate Do_2 with only fluids and vasopressors.

(c) Another multicenter, open-label study randomized ICU patients with a lactate concentration of 3 mmol/L or greater to the addition of lactate clearance of 20% or more (evaluated every 2 hours for the first 8 hours of therapy) or standard therapy (in which the treatment team was blinded for lactate concentrations).

(1) Both treatment arms had resuscitation targets of CVP 8–12 mm Hg, MAP 60 mm Hg or greater, and urinary output above 0.5 mL/kg/hour. $Scvo_2$ monitoring was optional in the standard therapy group but was mandated as part of the lactate clearance group. In the lactate clearance group, if the $Scvo_2$ was 70% or greater but the lactate concentrations did not decrease by 20%, vasodilators (e.g., nitroglycerin) were initiated to improve microvascular perfusion.

(2) Adding lactate clearance as a treatment goal was not associated with a difference in hospital mortality on bivariate analysis (33.9% vs. 43.5%, p=0.067). After adjustment for baseline differences between groups with multivariable analysis, a significant difference favored the lactate clearance group (HR 0.61; 95% CI, 0.43– 0.87; p=0.006).

(d) Given these findings, in a low-level recommendation, the SSC suggested targeting resuscitation to decrease lactate in patients with elevated lactate concentrations as a marker of tissue hypoperfusion.

(e) A recent multicenter trial, the ANDROMEDA-SHOCK trial, randomized patients with septic shock to either peripheral perfusion-targeted resuscitation (goal capillary refill time 3 seconds) or lactate level-targeted resuscitation (goal lactate normalization or a >20% reduction in lactate every 2 hours). This study found no association between resuscitation strategy and 28 day all cause mortality (HR 0.75; 95% CI, 0.55–1.02). This trial indicates that in resource-poor settings, or when lactate cannot be readily checked, capillary refill time may be a reasonable alternative to ensure adequate perfusion in patients with septic shock.

4. Blood pressure (MAP) goal
   a. As discussed previously, MAP is the true driving pressure for peripheral blood flow and end-organ perfusion and is preferred to SBP as a therapeutic target.
   b. A multicenter, open-label study randomized patients with septic shock to resuscitation with a MAP goal of either 65–70 mm Hg (low-target group) or 80–85 mm Hg (high-target group). The higher MAP target was achieved through vasopressor administration; patients in the high-target group had a significantly higher infusion rate and duration of vasopressors than did those in the low-target group, but the groups did not differ in total volume of fluid administration. The treatment arms did not differ in 28-day mortality (34.0% in the low-target group vs. 36.6% in the high-target group, p=0.57). However, the incidence of atrial fibrillation was significantly higher in the high-target group (6.7% vs. 2.8%, p=0.02). In an a priori–defined subgroup analysis of patients with chronic hypertension (with randomization stratified according to this covariate), those randomized to the high-target group had a lower incidence of a doubling of the SCr concentration (38.9% vs. 52.0%, p=0.02, stratum interaction p=0.009) and the need for renal replacement therapy (31.7% vs. 42.2%, p=0.046, stratum interaction p=0.04).
5. Summary and recommendations for initial resuscitation
   a. After an initial fluid challenge, fluid therapy should be continued, using a fluid challenge technique, until the patient is no longer fluid responsive.
   b. Vasopressors should be applied to initially target a MAP of 65–70 mm Hg, but the MAP goal may subsequently be adjusted if adequate organ perfusion is not attained (particularly in patients with chronic hypertension).
   c. Adequate tissue D_{O_2} should be ensured. If a central venous catheter is not inserted, lactate clearance is a reasonable target. If a central venous catheter is inserted, a combination of markers can be used (e.g., lactate clearance and Scv_{O_2} of 70% or greater).
6. Sepsis and septic shock care bundle
   a. The SSC, in collaboration with the Institute for Healthcare Improvement, has developed a core set of process steps and treatment goals grouped into a care bundle for patients with sepsis and septic shock.
   b. The goal of the care bundle is to improve early recognition and treatment of patients with sepsis and septic shock.
   c. This SSC care bundle was updated in 2015 in response to new evidence from the three previously noted quantitative resuscitation studies (Box 2).
   d. The updated bundle acknowledges the findings of the three studies and recommends using techniques in addition to CVP and Scv_{O_2} to reassess fluid responsiveness and tissue perfusion. These techniques include either use of a repeat focused examination by a licensed practitioner to evaluate for vital signs, cardiopulmonary findings, capillary refill, pulse, and skin findings or use of at least two of the following: CVP, Scv_{O_2}, bedside ultrasonography, or dynamic markers of fluid responsiveness (PLR or fluid challenge).
   e. This care bundle has been adopted by the Centers for Medicare & Medicaid Services as a quality (core) measure (SEP-1).

f. The slight differences between the SSC bundle and the SEP-1 quality measure are noted in Box 2.
g. In July 2016, a letter to the editor from the SEP-1 measure stewards and a representative from the Centers for Medicare & Medicaid Services indicated that the SEP-1 measure would not be updated to correspond with the new sepsis definitions until further data are available (JAMA 2016;316:457-8).
h. Adjustment of the SSC bundle and implementation of sepsis as a quality measure have led to a shift in resuscitation approaches for many clinicians.
i. Practitioners should systematically evaluate their institutional compliance and implement broad process steps to ensure compliance with the quality measure. These steps may include, but are not limited to, patient identification by clinical decision support tools in the electronic medical record and implementation of care paths and order sets for treatment.
j. In 2018, the SSC released an update recommending that all bundle elements be completed within 1 hour to ensure that aggressive resuscitation begins immediately in patients with septic shock. At this time, this recommendation has not been adopted by the Centers for Medicare & Medicaid Services.

Box 2. Sepsis and Septic Shock Management Bundle[a]

Accomplished within 3 hr of presentation[b]

1. Measure lactate concentration
2. Obtain blood cultures before administering antibiotics
3. Administer broad-spectrum antibiotics[c]
4. Administer crystalloid 30 mL/kg for hypotension or lactate ≥ 4 mmol/L[d]

If septic shock[e] is present, additional measures to be accomplished within 6 hr of presentation[b]

5. Apply vasopressors (for hypotension that does not respond to initial fluid resuscitation) to maintain MAP ≥ 65 mm Hg
6. If persistent arterial hypotension after initial fluid administration (MAP < 65 mm Hg) or if initial lactate was ≥ 4 mmol/L, reassess volume status and tissue perfusion, and document findings[f]
7. Re-measure lactate if the initial lactate concentration was elevated[g]

[a]Applies to all patients presenting with severe sepsis and septic shock (of note, the previous [Sepsis-2] definitions, which include the term *severe sepsis*, continue to be used for this bundle). Patients are excluded from the Centers for Medicare & Medicaid Services quality measure (SEP-1) for many reasons, including (1) they are transferred from another care facility (including an ED), (2) they have advanced directives for comfort care, (3) they have clinical conditions that preclude total measure completion (i.e., mortality within the first 6 hr of presentation), (4) they have a length of stay > 120 days, or (5) they were administered intravenous antibiotics within 24 hr of presentation. All components outlined must be fulfilled to satisfy the SEP-1 quality measure (it is an "all-or-none" measure).

[b]Time of presentation is defined as the time of triage in the ED or, if the patient is located in another care venue, from the earliest chart annotation consistent with all elements of severe sepsis or septic shock as ascertained through chart review.

[c]Before July 1, 2021, specific antibiotics were outlined in the SEP-1 quality measure to qualify as "broad spectrum." As of July 1, 2021, any IV antibiotic satisfies the measure if given within the appropriate time.

[d]Starting January 1, 2022, the SEP-1 quality measure allows for resuscitation volumes < 30 mL/kg if appropriate documentation with rationale is recorded in the medical record as outlined in the measure documents.

[e]Septic shock defined as hypotension (to SBP < 90 mm Hg, MAP < 70 mm Hg, or SBP decrease > 40 mm Hg from known baseline) or a lactate concentration ≥ 4 mmol/L.

[f]To meet the requirements, one of the following must be documented: (1) a focused examination by a licensed independent practitioner including vital signs, cardiopulmonary, capillary refill, pulse, and skin findings; or (2) any two of the following: measure CVP, measure $Scvo_2$, bedside cardiovascular ultrasonography, or dynamic assessment of fluid responsiveness with PLR or fluid challenge.

[g]In the SEP-1 quality measure, an initial lactate concentration ≥ 2 mmol/L is considered elevated.

Information from: Surviving Sepsis Campaign (SSC). Updated Bundles in Response to New Evidence [homepage on the Internet]. Available at www.survivingsepsis.org/SiteCollectionDocuments/SSC_Bundle.pdf; and The Joint Commission. Specifications Manual for National Hospital Inpatient Quality Measures. 2016. Available at https://www.jointcommission.org/specifications_manual_for_national_hospital_inpatient_quality_measures.aspx.

Patient Case

6. A 56-year-old woman with a medical history of hypertension presents to the ED with shortness of breath and cough productive of sputum. Her vital signs on admission are as follows: blood pressure 92/68 mm Hg, heart rate 104 beats/minute, respiratory rate 26 breaths/minute, and temperature 101.6°F (38.7°C). A chest radiograph reveals an opacity in the left lower lobe, but the radiograph is otherwise unremarkable. Her laboratory values of interest include Hgb 12.7 g/dL, WBC 16.4 × 10^3 cells/mm^3, Plt 80,000/mm^3, albumin 2.0 g/dL, lactate 3.2 mmol/L, and SCr 1.3 mg/dL. Her Glasgow Coma Scale score is 13. Which best describes the patient's condition?
   A. Systemic inflammatory response syndrome.
   B. Sepsis.
   C. Severe sepsis.
   D. Septic shock.

7. Antimicrobials
   a. Timing of initiation
      i. Adequate empiric antibiotics should be initiated within 1 hour after recognizing sepsis or septic shock.
      ii. A multicenter, retrospective study of patients with septic shock found that within the first 6 hours after the onset of hypotension, each hour of delay beyond the first hour in the administration of appropriate antibiotics was associated with a 7.6% decrease in hospital survival.
      iii. Other studies have further shown the importance of empiric antimicrobials used together with initial resuscitation, associating antimicrobial administration either before shock or within the initial hour of shock with improved survival.
   b. Initial empiric broad-spectrum therapy should include one or more drugs with activity against all likely pathogens (bacterial and/or fungal and/or viral).
      i. In an observational study of more than 5700 patients with septic shock, those who received initial appropriate antimicrobials had a significantly higher hospital survival rate than did those who received initial inappropriate antimicrobials (52.0% vs. 10.3%, p<0.0001).
      ii. Combination antibacterial therapy (at least two different classes of antibiotics) is indicated for patients with septic shock, but not for those with sepsis without shock.
         (a) A randomized controlled trial of patients with sepsis (termed *severe sepsis* in the study) that allocated patients to meropenem monotherapy or combination therapy with meropenem and moxifloxacin found no difference between groups in mean SOFA scores over 14 days (7.9 points vs. 8.3 points, p=0.36) or mortality rates at 28 or 90 days. Important caveats to this study are that the patient population studied was at a low risk of resistant pathogens (half of the patients had a community-acquired infection) and that moxifloxacin inadequately covers pathogens with a high likelihood of multidrug resistance (e.g., *Pseudomonas aeruginosa* and *Acinetobacter* spp.).
         (b) A meta-analysis that included 50 studies and more than 8500 patients with sepsis detected no overall mortality benefit of combination antibacterial therapy compared with monotherapy (pooled OR of death 0.86; 95% CI, 0.71–1.03, p=0.09); however, a stratified analysis showed significantly lower mortality with combination therapy in more severely ill patients (monotherapy risk of death greater than 25%, pooled OR of death with combination therapy 0.54; 95% CI, 0.45–0.66, p<0.001). The benefit of combination therapy was confined to patients with septic shock (with no benefit of combination therapy in patients without shock). In addition, a meta-regression analysis, which tried to elucidate

the differences in the benefit with combination therapy according to baseline mortality risk, showed a significant benefit of combination therapy in those with a mortality risk greater than 25%. These data suggest that severely ill patients benefit from combination antibacterial therapy.

iii. Combination therapy increases the likelihood that at least one drug is effective against the pathogen, particularly for known or suspected multidrug-resistant organisms such as *P. aeruginosa*.

iv. The 2021 SSC guidelines suggest combination therapy for patients at high risk of multidrug-resistant pathogens.

v. Empiric combination therapy should be de-escalated within the first few days of therapy if the patient has clinical improvement.

c. As noted in the Pharmacokinetics/Pharmacodynamics chapter, sepsis and septic shock can significantly affect the probability of attaining the antimicrobial pharmacokinetic/pharmacodynamic target. Dosing strategies for patients with sepsis should be optimized according to pharmacokinetic/pharmacodynamic principles.

i. Most notably, the volume of distribution of hydrophilic antibiotics (e.g., β-lactams, aminoglycosides, and vancomycin) will be increased. Clearance may either be increased (in the setting of augmented renal clearance) or decreased (in the presence of end-organ dysfunction).

ii. A pharmacokinetic/pharmacodynamic study of the first dose of β-lactams in patients with sepsis and septic shock suggested that the pharmacokinetic/pharmacodynamic target was attained in less than 50% of the patients given ceftazidime 2 g (28% target attainment), cefepime 2 g (16%), and piperacillin/tazobactam 4 g/0.5 g (44%). The pharmacokinetic/pharmacodynamic target was attained in 75% of patients receiving meropenem 1 g. For each antibiotic, the volume of distribution was higher and the clearance was lower than the values reported in healthy volunteers.

iii. A multicenter cross-sectional study of β-lactam concentrations in critically ill patients found that the minimum pharmacokinetic/pharmacodynamic target of at least 50% of free drug time above the minimal inhibitory concentration (MIC) (50% fT>MIC) was not achieved in 16% of patients with an infection. Of importance, achievement of 50% fT>MIC or 100% fT>MIC was independently associated with a higher likelihood of a positive outcome on multivariable analysis (OR 1.02; 95% CI, 1.01–1.04 and OR 1.56; 95% CI, 1.15–2.13, respectively).

iv. These data suggest that a loading dose approach for these antibiotics is necessary for patients with sepsis and septic shock. In addition, the impact of methods to improve the time that the free drug concentration is above the MIC should be studied further.

d. A recent retrospective analysis revealed that over 30% of patients had a delay in the second dose of antibiotics. Having a major delay in the second antibiotic dose was independently associated with increased odds of hospital mortality (OR 1.61; 95% CI, 1.01–2.57) and mechanical ventilation (OR 2.44; 95% CI, 1.24–4.69).

e. Antimicrobial therapy should be evaluated on a daily basis to determine whether opportunities for de-escalation or discontinuation exist.

i. Continued use of broad-spectrum antimicrobial therapy may cause untoward adverse effects and promote the development of resistance.

ii. De-escalation may be clear-cut in infections in which a contributive pathogen has been identified; in such cases, antimicrobial therapy should be reduced to the narrowest-spectrum agent with adequate activity. However, de-escalation may be more challenging in culture-negative sepsis.

iii. Antimicrobials are typically continued for 7–10 days, though longer courses may be indicated in patients with a poor clinical response, those with bacteremia, or those who are immunocompromised.

iv. Procalcitonin and other biomarkers may be used to limit the duration of antimicrobial therapy and are discussed in detail in the Infectious Diseases II chapter.

8. Source control measures (e.g., drainage of an abscess, debridement of infected necrotic tissue, or removal of a potentially infected device [including intravascular access devices]), as applicable, should be done as soon as possible (within 12 hours after the diagnosis is made).
9. Fluid therapy
   a. A fluid challenge technique should be used in which fluids are administered as long as the patient has improved clinical factors (particularly end-organ perfusion).
   b. After initial resuscitation, additional fluids should be guided by frequent reassessment of hemodynamic status. Dynamic markers (instead of static markers) of fluid responsiveness should be used, when available.
   c. Crystalloids (over colloids) are the recommended initial fluid type for an initial fluid challenge and subsequent intravascular replacement in patients with septic shock.
      i. Crystalloid solutions that approximate the electrolyte composition of plasma ("balanced" solutions) are an attractive alternative to 0.9% sodium chloride, and either may be used for fluid resuscitation (see section V, Agents Used to Treat Shock, for further discussion). The 2021 SSC guidelines suggest using balanced crystalloids instead of normal saline for resuscitation in patients with sepsis and septic shock.
         (a) No prospective randomized study has directly compared 0.9% sodium chloride with balanced salt solutions for fluid resuscitation when enrollment was restricted to patients with sepsis.
         (b) A propensity-matched retrospective cohort study of medical ICU patients with sepsis found that receipt of a balanced solution compared with 0.9% sodium chloride was associated with a lower incidence of in-hospital mortality (19.6% vs. 22.8%, p=0.001). Contrary to analyses of general critical care patients, this study found no difference in the incidence of acute renal failure between groups, which leads to questions regarding the mechanism of the detected mortality difference between groups.
         (c) A systematic review and network meta-analysis of patients with sepsis suggested a lower mortality rate in patients resuscitated with balanced solutions than in patients resuscitated with 0.9% sodium chloride (OR 0.78, credibility interval 0.58–1.05; low confidence in estimate of effect), but this difference was not statistically significant.
         (d) A recent meta-analysis of all trials comparing balanced crystalloids with 0.9% sodium chloride (including the aforementioned SMART study) in the patient cohort with sepsis found that mortality was not affected by fluid choice (OR 0.98; 95% CI, 0.80–1.20). However, major adverse kidney events (defined as the composite of death, new renal replacement therapy, or persistent renal dysfunction in the first 30 days) were reduced with balanced solutions (OR 0.78; 95% CI, 0.66–0.91).
   d. Although not recommended for initial resuscitation, iso-oncotic (4%–5%) albumin may be considered in some patients.
      i. A prospectively defined subgroup analysis of the SAFE study evaluated the effect of albumin 4% compared with 0.9% sodium chloride in patients with sepsis and septic shock. The unadjusted RR of death with albumin was 0.87 (95% CI, 0.74–1.02) in patients with sepsis and 1.05 (95% CI, 0.94–1.17) in patients without sepsis (p=0.06 for heterogeneity of treatment effect by subgroup). In a multivariable analysis that accounted for baseline factors, albumin administration was associated with a lower mortality risk (OR 0.71; 95% CI, 0.52–0.97; p=0.03).

ii. A systematic review and fixed-effect meta-analysis of albumin compared with alternative fluids for resuscitation in patients with sepsis found an association between albumin use and lower mortality (OR 0.82; 95% CI, 0.67–1.0; p=0.047). The benefit of albumin was retained when the analysis was restricted to crystalloids as the comparator (OR 0.78; 95% CI, 0.62–0.99; p=0.04). These data should be interpreted with caution, however, because many of the included studies had poor methodological quality, and when a random-effects model was used, the results for the overall analysis were not statistically significant (OR 0.84; 95% CI, 0.69–1.02, p=0.08).

iii. These data have renewed interest in 4%–5% albumin as a resuscitation fluid for patients with sepsis and septic shock. As such, the 2021 SSC guidelines suggest albumin as a component of the fluid resuscitation regimen when patients require a substantial amount of crystalloids.

e. Hydroxyethyl starch solutions should not be used for fluid resuscitation.

i. Compared with patients with sepsis allocated to lactated Ringer's solution, those allocated to pentastarch (a hydroxyethyl starch formulation) had a significantly higher incidence of acute renal failure (34.9% vs. 22.8%, p=0.002) and need for renal replacement therapy (31.0% vs. 18.8%, p=0.001), with no difference in 28-day mortality (26.7% vs. 24.1%, p=0.48).

ii. Compared with patients with sepsis randomized to receive Ringer's acetate solution, patients with sepsis randomized to receive hydroxyethyl starch had a significantly higher 90-day mortality (51% vs. 43%, p=0.03) and need for renal replacement therapy (22% vs. 16%, p=0.04).

iii. The increased need for renal replacement therapy and lack of mortality benefit in these studies led to the strong recommendation against the use of hydroxyethyl starch for fluid resuscitation in patients with sepsis and septic shock in the 2021 SSC guidelines.

f. Hyperoncotic (20%–25%) albumin replacement may be beneficial in patients with septic shock. An open-label study compared the replacement of albumin (with 20% albumin) to a goal serum albumin concentration of 3 g/dL plus crystalloid solution administration with the administration of crystalloid solution alone in patients with sepsis or septic shock. The albumin and crystalloid groups did not differ in the incidence of 28-day mortality (31.8% vs. 32.0%, p=0.94), but patients allocated to albumin had a shorter time to cessation of vasoactive agents (median 3 vs. 4 days, p=0.007). A post hoc subgroup analysis of patients with septic shock at enrollment showed that those randomized to albumin had a lower 90-day mortality rate (RR 0.87; 95% CI, 0.77–0.99); 90-day mortality did not differ in patients without septic shock (RR 1.13; 95% CI, 0.92–1.39; p=0.03 for heterogeneity). Of note, this post hoc subgroup analysis was performed on the secondary end point of 90-day mortality, and an analysis of the primary outcome of 28-day mortality was not reported. These data suggest that albumin replacement does not improve outcomes in patients with sepsis but that it has hemodynamic (and potentially mortality) advantages in patients with septic shock. Hence, the role of albumin replacement in patients with septic shock warrants further study.

10. Vasoactive agents and inotropes

a. Norepinephrine is the recommended first-line vasoactive medication. A meta-analysis of randomized trials that compared norepinephrine with dopamine for the treatment of septic shock found a higher risk of short-term mortality (RR 1.12; 95% CI, 1.01–1.20; p=0.035) and arrhythmias (RR 2.34; 95% CI, 1.46–3.77; p=0.001) in patients allocated to dopamine.

b. The optimal approach to using vasoactive agents and inotropes beyond the choice of initial vasopressor is unclear, but the choice should be based on patient-specific clinical factors.

i. Potential interventions could include using norepinephrine monotherapy (with appropriate dose escalation) or initiating an additional therapy (i.e., a second catecholamine vasopressor, arginine vasopressin (AVP), corticosteroids, an inotrope, or a combination of these therapies).

ii. The optimal time or norepinephrine dose at which to consider additional therapies is unknown. A dose that constitutes the failure of norepinephrine is not well defined in the literature, and the maximal doses used by clinicians (or institutions) are variable and often subjective.

c. Low-dose AVP (up to 0.03 units/minute) can be added to norepinephrine as the second-line adjunctive agent rather than escalating the dose of norepinephrine in patients not at goal MAP.
   i. In patients with septic shock, a relative vasopressin deficiency may exist, and patients may be sensitive to the vasoconstrictive effects of AVP.
   ii. Because of these effects, AVP has been used in patients with septic shock as both an endocrine replacement therapy (with a fixed-dose infusion) and a vasopressor (titrated to MAP).
   iii. The Vasopressin and Septic Shock Trial (VASST) compared the effects of adding AVP (0.01–0.03 units/minute) to norepinephrine with using norepinephrine monotherapy. Patients were initiated on the study drug an average of 12 hours after meeting the trial's inclusion criteria and when norepinephrine was at a mean dose of 20.7 mcg/minute. Overall, the study found no difference in 28-day mortality between treatment arms (35.4% vs. 39.3%, p=0.26), but norepinephrine requirements were significantly lower during the first 4 study days in the patients allocated to AVP (p<0.001). Adverse effects did not differ between groups. In an a priori–defined subgroup analysis of patients on the basis of shock severity (less severe shock defined as baseline norepinephrine dose 5–14 mcg/minute) with randomization stratified according to this covariate, 28-day mortality in the less severe shock stratum (defined as norepinephrine-equivalent doses less than 15 mcg/minute) was lower in patients randomized to AVP plus norepinephrine (26.5% vs. 35.7%, p=0.05), with no difference between groups in the more severe shock stratum (44.0% vs. 42.5%, p=0.76, stratum interaction p=0.10).
   iv. In the VANISH trial, which compared first-line vasopressin (up to 0.06 units/minute) with norepinephrine, there was no difference between groups in the incidence of kidney failure in survivors (absolute difference -2.3%; 95% CI, -13.0 to 8.5) or the number of kidney failure–free days in nonsurvivors (absolute difference -4 days; 95% CI, -11 to 5). Patients were initiated on the study drug within an average of 3.5 hours from shock onset and at a mean norepinephrine dose of 0.16 mcg/kg/minute. Patients were also randomized to hydrocortisone or placebo (in a 2 × 2 factorial design), but there was no significant interaction between vasopressin and hydrocortisone (interaction p=0.98 for 28-day mortality). Although the study intended to evaluate vasopressin as a first-line agent, 85% of patients were receiving open-label vasopressors before randomization.
   v. One consistent finding with the use of AVP in clinical trials is the reduction in norepinephrine dose requirements after its initiation, making it an attractive norepinephrine- and catecholamine-sparing agent.
   vi. Findings regarding the effects of AVP (and its analogs) on mortality have been conflicting in meta-analyses. One meta-analysis found a decreased mortality risk in patients with septic shock allocated to AVP (RR 0.87; 95% CI, 0.75–1.0; p=0.05), whereas another study found no significant benefit with AVP (RR 0.91; 95% CI, 0.79–1.05; p=0.21). Both meta-analyses found a significant benefit of AVP in decreasing norepinephrine doses. In an updated meta-analysis performed for the 2016 SSC guidelines, mortality did not differ between norepinephrine and vasopressin (RR 0.89; 95% CI, 0.79–1.00). A recent meta-analysis (which included the VANISH study) corroborated this and found no significant effect on mortality with vasopressin use (RR 0.98; 95% CI, 0.86–1.12).
   vii. The 2021 SSC guidelines note that the threshold for adding vasopressin is unclear, but it is reasonable to consider initiation when the dose of norepinephrine is 0.25–0.5 mcg/kg/minute.
   viii. High-dose vasopressin (doses above 0.03–0.04 units/minute) should be reserved for salvage therapy and, if used, should be used cautiously.

(a) A study of patients with vasodilatory shock (about 50% with septic shock) requiring high-dose norepinephrine (greater than 0.6 mcg/kg/minute) randomized patients to AVP 0.033 units/minute versus AVP 0.067 units/minute. The study was designed to evaluate hemodynamic changes, not mortality. Patients randomized to the higher dose had a greater reduction in norepinephrine requirements than did those allocated to the lower dose (p=0.006). High-dose AVP did not lead to a significantly lower cardiac index than low-dose AVP (as might be expected with this pure vasoconstrictor), but this finding is difficult to interpret because most patients were receiving concomitant inotropes, and the study was likely underpowered to assess this outcome.

(b) High-dose vasopressin is best reserved for patients with septic shock requiring high norepinephrine doses (greater than 0.6 mcg/kg/minute) with a high CO.

ix. Despite the unclear benefits of AVP on mortality, it offers an alternative mechanism to catecholamines for vasoconstriction, and low-dose AVP is commonly used in practice (often as the second vasoactive medication).

d. The SSC guidelines suggest adding epinephrine in patients with inadequate MAP levels despite norepinephrine and vasopressin.

i. A randomized trial of patients with septic shock compared epinephrine with norepinephrine with or without dobutamine. The groups did not differ in 28-day mortality (40% vs. 34%, p=0.31), but patients allocated to epinephrine had significantly higher lactate concentrations on day 1 (p=0.003) and lower arterial pH values on each of the first 4 study days. Caution should be used in concluding that mortality does not differ between epinephrine and norepinephrine because the study was powered to detect a 20% absolute difference in mortality rates, and a smaller difference between agents cannot be ruled out.

ii. The benefit of adding epinephrine to norepinephrine (whether this approach has a norepinephrine-sparing effect or whether it is best used in norepinephrine failure) is unclear.

iii. Because norepinephrine may cause tachycardia or tachyarrhythmias (often the impetus to limit doses), alternative vasopressors may be added. If catecholamine vasopressors with β_1-adrenergic properties (e.g., epinephrine) are added to augment MAP in patients receiving norepinephrine, they are unlikely to prevent tachyarrhythmias and will likely increase the risk of this adverse effect.

iv. In addition, epinephrine may preclude the use of lactate clearance as an initial resuscitation goal because it increases lactate concentrations through increased production by aerobic glycolysis (by stimulating skeletal muscle β_2-adrenergic receptors), an effect that likely wanes with continued epinephrine administration.

v. It seems most prudent to use epinephrine in patients receiving norepinephrine with a low MAP who require CO augmentation.

e. Phenylephrine is no longer recommended by the SSC guidelines for use in patients with septic shock because of limited clinical trial data.

i. Because of its afterload augmentation effects without β_1-adrenergic properties, phenylephrine may theoretically decrease SV and CO. As such, it is not recommended as a first-line vasopressor in patients with septic shock who have decreased CO because of inadequate preload.

ii. A small study (n=32) that randomized patients with septic shock to phenylephrine or norepinephrine as first-line therapy found no difference in hemodynamic measures (including CO) between agents in the first 12 hours of therapy. This finding is likely because of the inotropic effect of myocardial α_1-adrenergic receptor augmentation by phenylephrine. These data are in contrast to the theoretical concerns with phenylephrine and warrant further study.

iii. During a shortage of norepinephrine in 2011, phenylephrine was used as an alternative for patients with septic shock. Compared with normal use, in-hospital mortality was higher during the norepinephrine shortage (absolute risk increase of 3.7%; 95% CI, 1.5%–6.0%; p=0.03). Because phenylephrine was the most commonly used vasoactive agent during the shortage period, phenylephrine should be used cautiously in patients with circulatory shock.

iv. Because phenylephrine is a pure α_1-adrenergic agent, it is attractive for patients who develop a tachyarrhythmia while receiving norepinephrine. Although phenylephrine has been used in clinical practice for many years for this indication, no clinical trial data support this practice.

f. Dopamine is best used in a select patient population, including those with a low risk of tachyarrhythmias (which is difficult to predict) or those with bradycardia-induced hypotension. Low-dose ("renal dose" 2 mcg/kg/minute) dopamine should not be used to improve renal blood flow and urinary output (renal protection).

g. Preliminary data suggest that midodrine helps decrease vasopressor duration in patients in the convalescent phase of septic shock.

i. MIDAS was a multicenter, randomized, double-blind, placebo-controlled trial of patients with hypotension admitted to an ICU requiring a single-agent vasopressor who could not be liberated from vasopressors for at least 24 hours. Patients were randomized to receive midodrine 20 mg every 8 hours or placebo. There was no difference in time to vasopressor discontinuation in the 136 included patients (23.5 hours of midodrine vs. 22.5 hours of placebo). Of note, this study was not specific for patients with septic shock.

ii. Further, larger studies are needed before midodrine is used in routine practice.

iii. When midodrine is used for this indication, care should be taken to ensure appropriate discontinuation after resolution of shock. Reports have noted inappropriate continuation of midodrine that may have been found in the outpatient setting.

h. Dobutamine may be indicated in patients with evidence or suggestion of myocardial dysfunction.

i. A low CO is common in early septic shock but is often corrected with fluid resuscitation alone.

ii. Myocardial dysfunction may persist despite fluid resuscitation, with 39% of patients having left ventricular hypokinesia on ICU admission and an additional 21% of patients developing hypokinesia 24–48 hours after ICU admission in one series.

iii. If myocardial dysfunction leads to significantly impaired Do_2, an inotrope such as dobutamine (up to 20 mcg/kg/minute) may be indicated to improve CO.

iv. Dobutamine should not be used to achieve a predetermined supranormal cardiac index or $Scvo_2$.

i. As noted previously, levosimendan, compared with placebo, did not lead to a significant difference in the mean SOFA score in patients with sepsis. In a meta-analysis performed by the SSC, levosimendan and dobutamine did not differ with respect to mortality (RR 0.83; 95% CI, 0.66–1.05). Given these data, dobutamine is recommended as the preferred inotrope in patients with septic shock.

j. The 2021 SSC guidelines do not recommend angiotensin II as a first-line agent because of the low quality of evidence and lack of clinical experience in sepsis; however, they note that it may play a role as an adjunctive vasopressor therapy.

Patient Case

7. A 55-year-old man presents to the medical ICU with presumed urosepsis. His medical history is significant for congestive heart failure with a baseline ejection fraction of 20%. In the medical ICU, his vital signs are as follows: blood pressure 69/45 mm Hg, heart rate 84 beats/minute, respiratory rate 32 breaths/minute, and temperature 100.6°F (38.1°C). Blood cultures are obtained, and piperacillin/tazobactam is initiated. A central venous catheter shows CVP 18 mm Hg and $ScvO_2$ 70%. Which is the most appropriate initial vasopressor for this patient?
   A. Norepinephrine.
   B. Vasopressin.
   C. Dobutamine.
   D. Epinephrine.

11. Corticosteroids should only be used in patients with septic shock who do not achieve resuscitation goals despite fluid administration and vasopressors.
    a. Corticosteroids are an attractive option for patients with septic shock because of their anti-inflammatory effects (through inhibition of nuclear factor κB) and ability to improve blood pressure response to catecholamines (through up-regulation of adrenergic receptors and potentiation of vasoconstrictor actions).
    b. Use of corticosteroids for patients with septic shock has been a source of controversy for years.
    c. In studies of short courses of high-dose corticosteroids (typically at doses of 30 mg/kg of methylprednisolone or greater) in patients with sepsis and septic shock, corticosteroids did not improve patient outcomes.
    d. Four large studies have evaluated the effect of corticosteroids in patients with septic shock, with conflicting findings on mortality.
        i. In a French trial of patients with septic shock and vasopressor-unresponsive shock (i.e., inability to increase SBP above 90 mm Hg for 1 hour despite fluids and vasopressors), patients randomized to low-dose hydrocortisone and fludrocortisone had improved survival in a time-to-event analysis (HR 0.71; 95% CI, 0.53–0.97). The mortality benefit with corticosteroids was limited to patients unable to increase their cortisol concentration by more than 9 mcg/dL in response to ACTH administration (nonresponders).
        ii. In a larger, multicenter Corticosteroid Therapy of Septic Shock (CORTICUS) study, which had less-stringent inclusion criteria than the earlier-noted French trial, hydrocortisone administration was not associated with improved survival in ACTH nonresponders (28-day mortality 39.2% vs. 36.1%, p=0.69).
        iii. The Adjunctive Corticosteroid Treatment in Critically Ill Patients with Septic Shock (ADRENAL) trial randomized 3800 patients to hydrocortisone or placebo. 90-day mortality did not differ between the hydrocortisone or placebo groups (27.9% vs. 28.8%, OR 0.95; 95% CI, 0.82–1.10; p=0.50). However, patients randomized to hydrocortisone had quicker time to shock resolution (3 vs. 4 days; p<0.001). A subgroup analysis of patients who received hydrocortisone on the basis of time from shock onset to randomization showed improvement in death at 90 days in patients who received hydrocortisone within 6–12 hours (OR 0.71; 95% CI, 0.54–0.94).

iv. The Activated Protein C and Corticosteroids for Human Septic Shock (APROCCHSS) trial randomized 1241 patients to hydrocortisone and fludrocortisone or placebo within 24 hours of shock onset. Patients were also randomized to drotrecogin alfa (activated) in a 2 x 2 factorial design, but after the drug was withdrawn from the market, the study was continued with only the corticosteroid randomization. Patients in the hydrocortisone and fludrocortisone group had lower 90-day mortality than those receiving placebo (43.0% vs. 49.1%; RR 0.88; 95% CI, 0.78–0.99; p=0.03). In addition, patients randomized to the corticosteroid arm had increased vasopressor-free days (16 vs. 11 days; p<0.001) and organ failure–free days (14 vs. 13 days; p=0.003).

v. These studies differed in the patient populations evaluated, with patients more severely ill demonstrating a mortality benefit from corticosteroids. When this question was specifically evaluated in the ADRENAL trial, there was no heterogeneity of effect by subgroup, suggesting severity of illness did not influence the outcome.

vi. Other notable differences between studies that may account for the disparate results are the drug regimens used and the timing of therapy initiation.

e. Two meta-analyses that included these four studies found that corticosteroids had no mortality benefit (RR 0.96; 95% CI, 0.91–1.02 in one meta-analysis and RR 0.93; 95% CI 0.84–1.03 in another meta-analysis).

f. In meta-analyses, corticosteroid administration has consistently been associated with improvements in shock reversal and a higher incidence of adverse effects (particularly hypernatremia and hyperglycemia).

g. A recent evaluation using the Premier Healthcare Database evaluated the addition of fludrocortisone to hydrocortisone compared with hydrocortisone alone in patients with septic shock and found lower rates of the primary composite outcome of in-hospital death or discharge to hospice (47.2% vs. 50.8%; adjusted risk difference: –3.7% 95% confidence –4.2% to –3.1%) in the group of patients who received fludrocortisone. Additional data are needed to confirm the findings from this retrospective evaluation and definitively determine if the addition of fludrocortisone to the standard regimen of hydrocortisone is necessary.

h. Guidelines from the Society of Critical Care Medicine and the European Society of Intensive Care Medicine suggest using corticosteroids in patients with septic shock that is not responsive to fluids and moderate- to high-dose vasopressor therapy. These guidelines further state that if hydrocortisone is indicated, a dose of less than 400 mg/day for 3 days or more at full dose is suggested.

i. The SSC guidelines suggest using hydrocortisone in patients with septic shock and an ongoing requirement for vasopressor therapy at a dose of 200 mg/day given as either 50 mg intravenously every 6 hours or a continuous infusion. The guidelines note the suggestion that hydrocortisone be initiated when norepinephrine or epinephrine doses exceed 0.25 mcg/kg/minute at least 4 hours after initiation.

j. Although two of the randomized controlled trials mentioned earlier used fludrocortisone, the SSC guidelines recommend hydrocortisone alone. This is likely because of the COIITSS study, a 2 x 2 factorial trial that compared hydrocortisone in combination with fludrocortisone with hydrocortisone alone. No difference in the primary outcome of in-hospital death (42.9% hydrocortisone and fludrocortisone vs. 45.8% hydrocortisone alone) or other clinical outcomes was detected between treatment arms.

k. Because of immunoassay imprecision and the inconsistent benefit of identifying patient response, the ACTH test should not be used to identify patients for corticosteroid administration.

Patient Case

8. A 62-year-old woman was admitted to the surgical ICU with presumed aspiration pneumonia. The patient has a history of insulin-dependent diabetes and hyperthyroidism. On admission, she was intubated and required aggressive resuscitation and vasopressor therapy. Currently, the patient is receiving norepinephrine 8 mcg/minute and has the following vital signs and laboratory values: blood pressure 100/60 mm Hg, heart rate 90 beats/minute, and CVP 14 mm Hg; her $ScvO_2$ is 55%. Which is best regarding the patient's steroid therapy?
   A. Administer hydrocortisone 50 mg every 6 hours.
   B. Administer hydrocortisone 8.3 mg/hour as a continuous infusion.
   C. Perform the ACTH stimulation test.
   D. No steroids are necessary right now.

12. Supportive therapies
    a. Vitamin C, hydrocortisone, and thiamine
        i. A retrospective before-after study evaluated the effect of the combination of intravenous vitamin C (1.5 g every 6 hours for 4 days or until ICU discharge), hydrocortisone (50 mg every 6 hours for 7 days or until ICU discharge, followed by a taper over 3 days), and intravenous thiamine (200 mg every 12 hours for 4 days or until ICU discharge) for patients with sepsis. Patients receiving this medication combination had a lower propensity-adjusted odds of mortality (OR 0.13; 95% CI, 0.04–0.48; p=0.002). This study is limited by the before-after design and a control group mortality rate (8.5%) lower than in contemporary sepsis studies (leading to questions about the study population).
        ii. Several subsequent randomized controlled trials have evaluated this clinical question, including the VITAMINS, VICTAS, and ACTS trials, which have all shown no benefit with the use of vitamin C, hydrocortisone, and thiamine.
        iii. Two randomized controlled trials, the CITRIS-ALI and the LOVIT trials, evaluated vitamin C without hydrocortisone or thiamine in patients with sepsis and found no clinical benefit to the use of vitamin C. In fact, the LOVIT trial found higher rates of death or persistent organ dysfunction at 28 days in the vitamin C cohort (44.5% vs. 38.5%; RR 1.21; 95% CI, 1.04−1.40).
        iv. The 2021 SSC guidelines suggest against the use of intravenous vitamin C in patients with sepsis and septic shock. They do not discuss the role of thiamine or the combination of vitamin C, hydrocortisone, and thiamine.
    b. Intravenous immunoglobulins should not be used in patients with sepsis and septic shock. In a randomized study of patients with sepsis, 28-day mortality was no different between patients allocated to intravenous immunoglobulin G and placebo (39.3% vs. 37.3%, p=0.67).
    c. A transfusion threshold of 7 g/dL or less is appropriate for patients with septic shock unless extenuating circumstances exist (i.e., myocardial ischemia, severe hypoxemia, or acute hemorrhage).
        i. A multicenter study compared two different transfusion thresholds in patients with septic shock. The study enrolled patients with septic shock and a hemoglobin concentration below 9 g/dL to receive PRBC transfusion if their hemoglobin was 7 g/dL or less (lower threshold) or 9 g/dL or less during their ICU stay.
        ii. As expected, patients allocated to the lower-threshold group less commonly received PRBC transfusion and had lower hemoglobin concentrations.
        iii. Treatment arms did not differ in 90-day mortality (43.0% in the lower threshold group vs. 36.6% in the higher-threshold group, p=0.44). Patients in the lower-threshold group had no higher incidence of ischemic events in the ICU (7.2% vs. 8.0%, p=0.64), and outcomes did not differ between groups in the subgroup of patients with chronic cardiovascular disease (RR 1.08 [95% CI, 0.75–1.40]; p=0.25 for heterogeneity of treatment effect by subgroup; only 14% of the overall population had chronic cardiovascular disease).

iv. Although the study did not specifically evaluate the practice of transfusing blood in patients with evidence of hypoperfusion (low $ScvO_2$ or elevated lactate concentrations) associated with a low hemoglobin concentration, the median $ScvO_2$ at baseline was below 70% and lactate concentrations were above 2 mmol/L in both groups, suggesting that most patients had evidence of hypoperfusion.
v. These data suggest that use of a hemoglobin transfusion threshold of 7 g/dL or lower is safe for most patients, including those with sepsis and septic shock.

13. Continued care of the patient with septic shock (resuscitation end points beyond first 6 hours): See section IV, Resuscitation Parameters and End Points, for an additional discussion of this topic.
    a. Uncertainty persists regarding hemodynamic targets beyond the first 6 hours of presentation.
        i. Therapy should be directed toward maintaining adequate end-organ perfusion and normalization of lactate concentrations, but the specific method(s) to achieve these goals will be patient-specific.
        ii. A strategy of systematically increasing CO to predefined "supranormal" values was not associated with a mortality benefit; hence, it is not recommended.
    b. Care should be used to avoid giving excessive fluids.
        i. In a retrospective analysis of data from a randomized controlled trial of patients with septic shock, patients in the highest quartile of fluid balance had a significantly higher mortality rate than patients in the lowest two quartiles. This association was present when fluid balance was evaluated at both 12 hours and 4 days after study enrollment. In the same analysis, a CVP greater than 12 mm Hg conferred a higher risk of mortality than a lower CVP.
        ii. A meta-analysis of 9 randomized controlled trials (n=637) that compared interventions targeted at limiting fluid administration in patients with sepsis found no difference in all-cause mortality (RR 0.87; 95% CI, 0.69−1.10) or serious adverse effects (RR 0.91; 95% CI, 0.78−1.05) with the use of lower versus higher fluid volumes.
        iii. Despite conflicting data between prospective (CLOVERS and CLASSIC) and retrospective studies, fluid administration should only be given to patients who are proven or predicted to respond to fluid. This is best accomplished using dynamic markers of fluid responsiveness.
    c. Resuscitation targeted to improving microcirculatory perfusion is a potential new therapeutic frontier.
        i. Studies have shown that the microcirculation is often altered in patients with sepsis, persistent microvascular alterations are associated with multisystem organ failure and death, alterations are more severe in non-survivors than in survivors, and improvements in microcirculatory blood flow correspond with improved patient outcomes.
        ii. Impaired microcirculatory perfusion may at least partly explain why patients have elevated lactate concentrations despite achievement of (macrovascular) hemodynamic goals.
        iii. Several strategies to improve microcirculatory perfusion have been investigated.
            (a) In a nonrandomized trial, fluid resuscitation improved microvascular perfusion in early, but not late, sepsis. Another study found that microcirculatory perfusion improved with PLR or fluid administration in patients who were fluid responsive.
            (b) Use of norepinephrine to increase the MAP above 65 mm Hg has not been associated with correction of impaired microcirculatory perfusion.
            (c) In a randomized trial of patients who underwent quantitative resuscitation, adding nitroglycerin was not associated with improved microcirculatory blood flow. In-hospital mortality was numerically higher in patients allocated to nitroglycerin than in placebo, but this difference was not statistically significant (34.3% vs. 14.2%, p=0.09).
            (d) A prospective open-label study evaluated the effects of dobutamine 5 mcg/kg/minute initiated after quantitative resuscitation (but within the first 48 hours of presentation). Dobutamine significantly improved microcirculatory perfusion compared with baseline. Of interest, the beneficial effects of dobutamine were unrelated to changes in cardiac index or blood pressure.

(e) Initiation of inhaled nitric oxide after quantitative resuscitation did not improve microcirculatory flow or lactate clearance.
(f) In summary, fluid resuscitation and dobutamine initiation appear to improve microcirculatory perfusion. It is still unknown whether using these therapies to improve microcirculatory perfusion can improve patient outcomes such as mortality.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: B**

The patient has hypovolemic shock caused by his upper GI hemorrhage and has symptoms of compromised end-organ perfusion (i.e., confusion). Although his history of hypertension is relevant in relation to determining a resuscitation goal, it does not acutely affect DO_2 (Answer A is incorrect). Tachycardia is a symptom in response to his hypovolemia, and, in the absence of complicating factors (e.g., atrial fibrillation or LV diastolic dysfunction), tachycardia will increase (not decrease) DO_2 (Answer C is incorrect). Leukocytosis does not impede DO_2 until it reaches an exorbitant threshold (greater than 75 L/mm^3) (Answer D is incorrect). When examining the Fick equation for DO_2, acute anemia is the determinant that is adversely affecting DO_2 (Answer B is correct).

2. **Answer: C**

The patient has evidence of vasodilatory shock post-resuscitation, with his relatively high CVP and PCWP values and high CO (Answer C is correct). This is further confirmed by a calculated SVR of 541 dynes x second x cm^{-5}. The patient could be thought to have spontaneous bacterial peritonitis in the setting of cirrhosis and ascites complicated by upper GI hemorrhage. Even though his presentation suggests hypovolemic shock (from GI hemorrhage), his MAP did not respond to fluid and blood product administration. Furthermore, he does not have low preload or low CO (Answer A is incorrect). If the patient had a low CO together with a high SVR, cardiogenic shock or obstructive shock might be possible (with one differentiation depending on CVP/right atrial pressure and PCWP), but this is untrue for the patient (Answers B and D are incorrect).

3. **Answer: C**

The patient's clinical scenario of refractory hypoxemia and hypotension with hypoperfusion suggests that an accurate prediction of fluid responsiveness is needed. Dynamic markers of fluid responsiveness (e.g., SVV) are superior to static markers of fluid responsiveness (e.g., CVP and PCWP) (Answer C is correct; Answers A and B are incorrect). A low MAP may be from either a low CO or a low SVR. Furthermore, a low preload is only one of many components that may contribute to a low CO. As such, a low MAP is not a good predictor of fluid responsiveness (Answer D is incorrect).

4. **Answer: C**

The patient has shock with hypoperfusion and a positive response to a PLR test, which suggests he is still fluid responsive (Answer D is incorrect). Data from a large randomized trial of patients with heterogeneous shock types showed no difference in efficacy and safety between albumin and 0.9% sodium chloride, but the cost of albumin is substantially higher. These data suggest that crystalloids such as 0.9% sodium chloride are preferred for fluid resuscitation in the ICU. In addition, the patient has not received a substantial volume of fluid (he has received less than 30 mL/kg; Answer C is correct; Answer A is incorrect). Hydroxyethyl starch, which has been associated with an increased need for renal replacement therapy without a mortality benefit, should be avoided for fluid resuscitation in the ICU (Answer B is incorrect).

5. **Answer: D**

The patient has evidence of ventricular dysfunction with a low $ScvO_2$ and poor ventricular contractility on echocardiogram. A vasoactive agent with strong inotropic properties is indicated. Epinephrine has strong β_1-adrenergic properties and is the best selection in this case (Answer D is correct). Both phenylephrine and vasopressin are essentially pure vasoconstrictors that do not increase CO and could theoretically decrease CO. In addition, use of phenylephrine is no longer recommended by the guidelines (Answers A and B are incorrect). Although norepinephrine has β_1-adrenergic properties, it primarily increases blood pressure through vasoconstriction secondary to its α_1-adrenergic properties, with only minimal effects on CO. Increasing norepinephrine in this case would probably not improve the patient's CO sufficiently to improve tissue perfusion (Answer C is incorrect).

6. **Answer: B**

The patient has life-threatening organ dysfunction secondary to infection. Although not all the criteria to calculate the score are available, her SOFA score is at least 4 points (2 points for Plt, 1 point for Glasgow Coma Scale score, and 1 point for SCr), indicating organ dysfunction. In addition, the patient fulfills all three qSOFA criteria (altered mental status, SBP of 100 mm Hg or less, and respiratory rate of 22 breaths/minute or greater).

Therefore, she meets the criteria for sepsis, according to the new definition (Answer B is correct). Although she has an elevated lactate concentration, she does not require vasopressors to maintain a MAP above 65 mm Hg and therefore does not meet the criteria for septic shock (Answer C is incorrect). In the new sepsis definition, severe sepsis and systemic inflammatory response syndrome are no longer clinical entities in the spectrum of sepsis severity and should not be used (Answers A and D are incorrect).

7. **Answer: A**

The patient currently has a MAP less than 65 mm Hg and signs of a global lack of perfusion with an increased lactate concentration. Vasopressor therapy is indicated to sustain perfusion. This patient has underlying severe congestive heart failure with an ejection fraction of 20%; however, the $Scvo_2$ shows that the patient's Do_2 is sufficient, which suggests he is in distributive septic shock as opposed to cardiogenic shock. The SSC guidelines recommend initiating vasopressor therapy to target a MAP of 65 mm Hg and norepinephrine as the first-choice vasopressor (Answer A is correct). Vasopressin is currently recommended only as a secondary vasopressor, as an addition to catecholamine therapy (Answer B is incorrect). In septic shock, dobutamine is recommended only when the $Scvo_2$ is less than 70% and the patient has an adequate Hgb concentration (Answer C is incorrect). Epinephrine is currently recommended as an alternative to norepinephrine. In a study that compared epinephrine with norepinephrine plus dobutamine, the two arms did not differ in mortality outcomes; however, epinephrine was associated with lower pH values and higher lactate concentrations on day 1 (Answer D is incorrect).

8. **Answer: D**

According to the SSC guidelines, corticosteroids may be considered if patients have a poor response to fluid resuscitation and vasopressor therapy. In addition, guidelines from the Society of Critical Care Medicine and the European Society of Intensive Care Medicine suggest corticosteroids only in patients requiring moderate- to high-dose vasopressors. In this case, the patient is receiving relatively low doses of vasopressors with an adequate MAP; hence, no steroids are currently necessary (Answer D is correct; Answers A and B are incorrect). Hydrocortisone is recommended at a dose of 200–400 mg/day. Administering it as a continuous infusion could be considered because this might lead to fewer variations in serum glucose. The ACTH stimulation test to determine which patients with sepsis should receive hydrocortisone is no longer recommended because the CORTICUS study showed that the ACTH stimulation test did not predict response to hydrocortisone therapy (Answer C is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**
This older adult patient presents with septic shock. Shock is a syndrome of impaired Do_2, leading to tissue injury and end-organ failure. In this case, it is important to realize that fever is a symptom of an inflammatory syndrome, not a shock syndrome (Answers A, B, and D are incorrect). This patient's presentation with hypotension, confusion, and hyperlactatemia suggests the presence of a shock syndrome in the setting of impaired Do_2 (Answer C is correct).

2. **Answer: B**
The Fick equation for Do_2 best shows Do_2. According to this equation, Do_2 depends on CO and Cao_2. Cardiac output depends on heart rate and SV. Typically, an elevated heart rate will increase CO and Do_2. However, if a patient has atrial fibrillation, ventricular filling is impaired and SV is decreased, causing a decrease in CO and Do_2 (Answer B is correct). Lactate does not impede Do_2 but is a byproduct of impaired Do_2 (Answer A is incorrect). Similarly, Do_2 is not impaired by acute kidney injury (Answer C is incorrect). Finally, fever is an inflammatory response that increases Vo_2 but does not impair Do_2 (Answer D is incorrect).

3. **Answer: D**
The patient's laboratory values and arterial pH are consistent with hyperchloremic metabolic acidosis, and chloride-rich fluids should be avoided. Lactated Ringer's solution is considered a relatively chloride-poor solution (chloride content 111 mEq/L) and is best in this case (Answer D is correct). With its chloride content of 154 mEq/L, 0.9% sodium chloride is considered a chloride-rich fluid. Liberal use of chloride-rich fluids such as 0.9% sodium chloride has been associated with an increased need for renal replacement therapy (Answer A is incorrect). Although albumin 5% is considered a chloride-poor solution, data from randomized controlled trials have not supported a mortality benefit with albumin over crystalloids, even in the setting of hypoalbuminemia. Moreover, the patient has not received a "substantial volume" of crystalloids (she has received less than 30 mL/kg of fluid) that would coincide with recommendations to give albumin (Answer B is incorrect). Hydroxyethyl starch solutions may also have a high chloride content (depending on the formulation) and have been associated with an increased need for renal replacement therapy in the general critical care population, with no mortality benefit. Hydroxyethyl starch solutions should be avoided for fluid resuscitation in the ICU (Answer C is incorrect).

4. **Answer: B**
The patient has evidence of end-organ hypoperfusion (urinary output less than 0.5 mL/kg/hour and elevated lactate concentration without significant clearance), despite a MAP greater than 65 mm Hg and quantitative resuscitation. The patient probably needs a higher perfusion pressure because of a right-shifted zone of autoregulation secondary to hypertension. The norepinephrine dose should be increased to target a higher MAP; the exact target will depend on the patient's response and should be selected as the threshold that improves end-organ perfusion (Answer B is correct). The patient has no evidence of impaired CO (his $Scvo_2$ is not low); therefore, fluids (to improve SV) and inotropes (e.g., dobutamine) are not indicated (Answers C and D are incorrect). Because the patient has continued evidence of hypoperfusion, action should be taken, and the current therapy should be modified (Answer A is incorrect).

5. **Answer: B**
The patient has hypotension and signs of hypoperfusion with an elevated lactate concentration; possible interventions such as fluid administration should be explored further. Because the patient requires a high Fio_2, a reliable predictor of fluid responsiveness should be used to guide fluid therapy instead of administering fluid without respect to predicting responsiveness. In the setting of atrial fibrillation, an elevated PPV is not a reliable predictor of fluid responsiveness, and further evaluation should be done before fluids are administered (Answer A is incorrect). A PLR test can be used in both spontaneously breathing patients and those receiving mechanical ventilation to predict fluid responsiveness, and the patient has a method to assess the presence (or absence) of a CO response (Answer B is correct). Although the accuracy of the CO value from arterial pulse pressure waveform analysis may be somewhat limited by atrial fibrillation, this may be accounted for with the internal software of most devices and can be used to gauge a response to the PLR test. The patient has a femoral central venous catheter, which cannot be used to assess hemodynamic markers such as CVP or $Scvo_2$ (Answers C and D are incorrect). Furthermore, CVP is an inadequate predictor of fluid responsiveness.

6. **Answer: D**
The patient has features of vasodilatory shock secondary to an immune-mediated ("anaphylactic") reaction (low preload, a low $Scvo_2$ [suggestive of poor Do_2], and an elevated lactate concentration). The patient should receive aggressive fluid resuscitation and be initiated on a vasopressor such as norepinephrine with the primary effects of augmenting afterload (Answer D is correct). Although the patient has features of poor Do_2, this is likely because of inadequate preload, which will be augmented by fluid administration. Agents targeted toward improving Cao_2 (PRBCs) and CO (dobutamine and milrinone) should not be initiated unless the patient has inadequate Do_2 and is not fluid responsive (Answers A–C are incorrect).

7. **Answer: A**
The patient has received an initial fluid challenge of only 23 mL/kg of crystalloids and still has evidence of end-organ hypoperfusion (elevated lactate concentration and urinary output less than 0.5 mL/kg/hour). An additional bolus of at least 500 mL of 0.9% sodium chloride is indicated to ensure an initial fluid challenge of at least 30 mL/kg of crystalloids and to improve end-organ perfusion (Answer A is correct). Because the patient has not received a complete initial fluid challenge (or even a substantial amount of crystalloids), albumin is not indicated (Answer B is incorrect). Vasopressors are not indicated right now because the patient's MAP is above 65 mm Hg (the patient's MAP is 67 mm Hg). In addition, if an initial vasopressor were to be selected, norepinephrine would be preferred (Answer C is incorrect). The patient's low $Scvo_2$ is likely caused by inadequate preload (resulting in inadequate SV and CO). Adequate preload should be ensured before giving PRBCs as part of improving Do_2 (Answer D is incorrect).

8. **Answer: A**
Despite an initial fluid challenge of greater than 30 mL/kg of crystalloids, the patient has continued evidence of hypotension (MAP 63 mm Hg) and hypoperfusion (an elevated lactate and a urinary output less than 0.5 mL/kg/hour). A vasopressor should be initiated to improve blood pressure (MAP greater than 65 mm Hg) and organ perfusion. Norepinephrine is recommended by the SSC as the first-line vasopressor (Answer A is correct). Vasopressin is not recommended as the single initial vasopressor, but it may be added to norepinephrine (Answer B is incorrect). Phenylephrine is no longer recommended in the SSC guidelines. Although this patient has a history of atrial fibrillation and a high heart rate, norepinephrine should still be tried and the patient observed for signs of worsening tachyarrhythmias (Answer C is incorrect). Dopamine is recommended as an alternative vasopressor to norepinephrine in select patients, such as those with bradycardia. In a meta-analysis of patients with septic shock, dopamine was associated with a higher mortality rate and more frequent tachyarrhythmias (Answer D is incorrect).

9. **Answer: B**
Vasopressin can be added to norepinephrine to either increase MAP or lower norepinephrine requirements. In the VASST (Vasopressin and Septic Shock Trial) study comparing norepinephrine monotherapy with norepinephrine plus vasopressin, mortality did not differ between the two groups, but norepinephrine requirements were significantly lower in the patients allocated to receive AVP. According to the SSC guidelines, vasopressin can be added to norepinephrine to either raise the MAP or decrease the norepinephrine dosage (Answer B is correct). Although a dynamic marker of fluid responsiveness is not presented, the patient's CO and cardiac preload are likely adequate, given that his $Scvo_2$ is 72%. Additional fluid loading is not indicated, given the information presented (Answer A is incorrect). Phenylephrine is no longer recommended as a treatment in the SSC guidelines; however, it has theoretical benefit as a second-line vasopressor when a malignant tachyarrhythmia is associated with norepinephrine. In this case, phenylephrine is not indicated because sinus tachycardia is not considered a malignant tachyarrhythmia (Answer C is incorrect). Epinephrine is recommended when an additional agent is needed to raise the MAP to the target. Adding epinephrine to norepinephrine would only add inotropic support, but this patient has no signs of low CO with an $Scvo_2$ of 72%, and the patient's inadequate blood pressure is the most likely reason for his hypoperfusion (elevated lactate and low urinary output). Adding epinephrine would also likely increase the patient's heart rate and potential for a tachyarrhythmia. As such, epinephrine is not indicated (Answer D is incorrect).